LOGTEL
It's more than knowledge

IOT (and M2M and WoT)

Day 2 - Technologies | 2026

By Samuel Dratwa
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The course: four days, one story

Day 1
Foundations

- Networking: OSI and TCP/IP
- Wi-Fi
- Wireshark: seeing packets
- Hardware interfaces
- What IoT is - and why now
- Bluetooth, NFC, RFID, UWB

Day 2
Technologies

- Architecture and cloud
- LPWAN, cellular, satellite
- LoRa and LoRaWAN
- Zigbee, Thread, Matter
- MQTT and brokers
- Grafana dashboards
- MakeCode warm-up

TODAY

Day 3
Cellular IoT and security

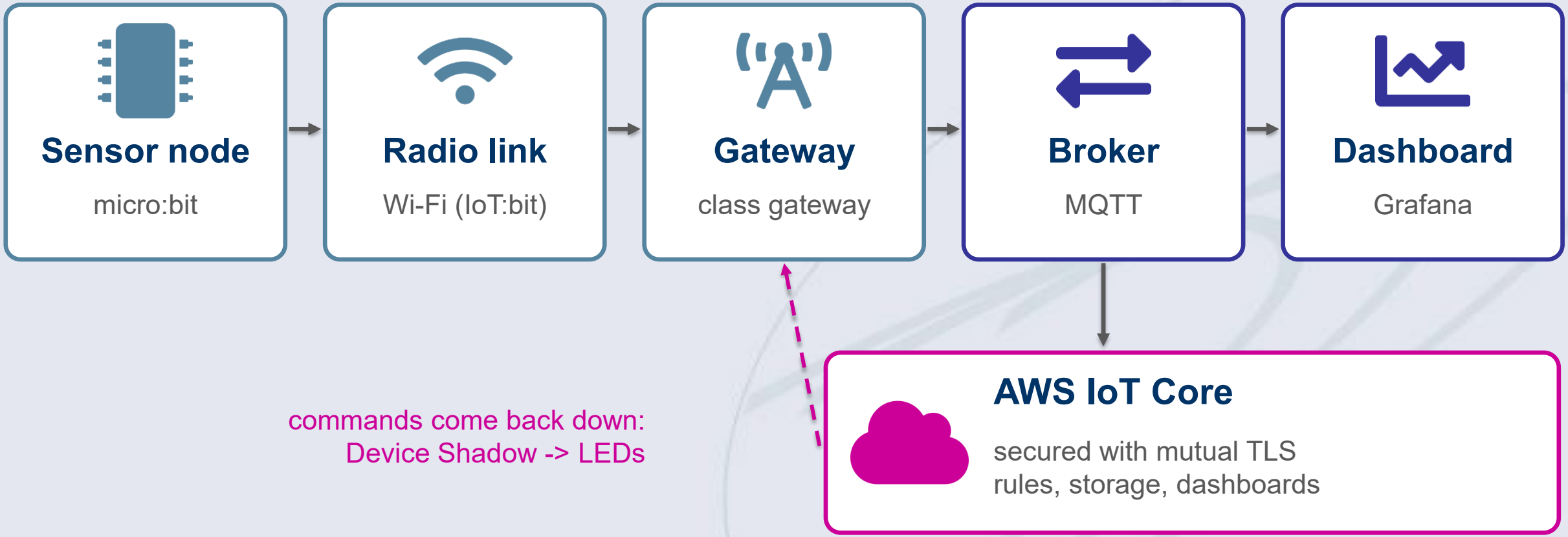
- Cellular IoT: NB-IoT, LTE-M
- D2D and V2X
- Device management, OTA updates
- End-to-end IoT security
- Cellular modem practice

Gilad

Day 4
Hands-on lab

- AWS IoT Core
- Certificates and mTLS
- micro:bit to the cloud
- MQTT + Wireshark + Grafana
- Cloud-to-device control
- Troubleshooting challenges

The pipeline we build - one step per day



Day 1 - understand each part

Day 2 - messaging, dashboards

Day 4 - build it end to end

Day 2 - Technologies: today's plan

1 

Architecture and cloud

reference models, edge, platforms

2 

LPWAN, cellular, satellite

the long-range options

3 

LoRa and LoRaWAN

the core of today

4 

Zigbee, Thread, Matter

mesh inside buildings

5 

MQTT

publish / subscribe messaging

6 

Grafana

time series and dashboards

7 

MakeCode warm-up

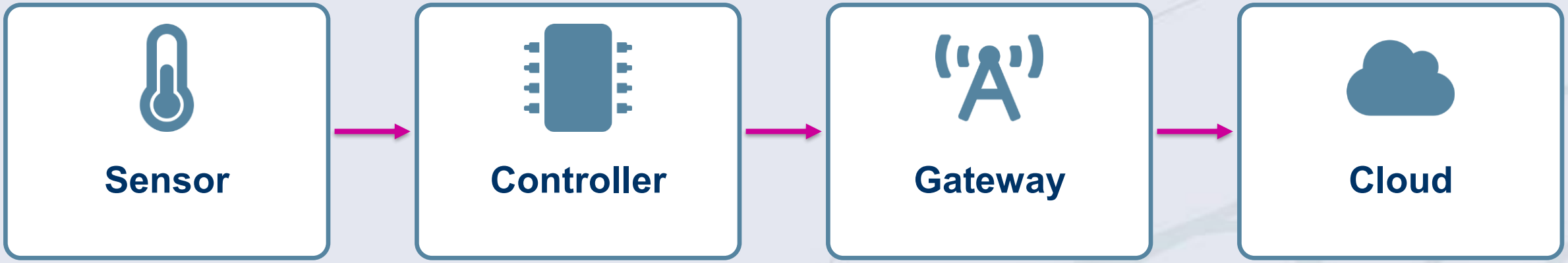
30 minutes with the micro:bit

8 

Summary and quiz

what to remember

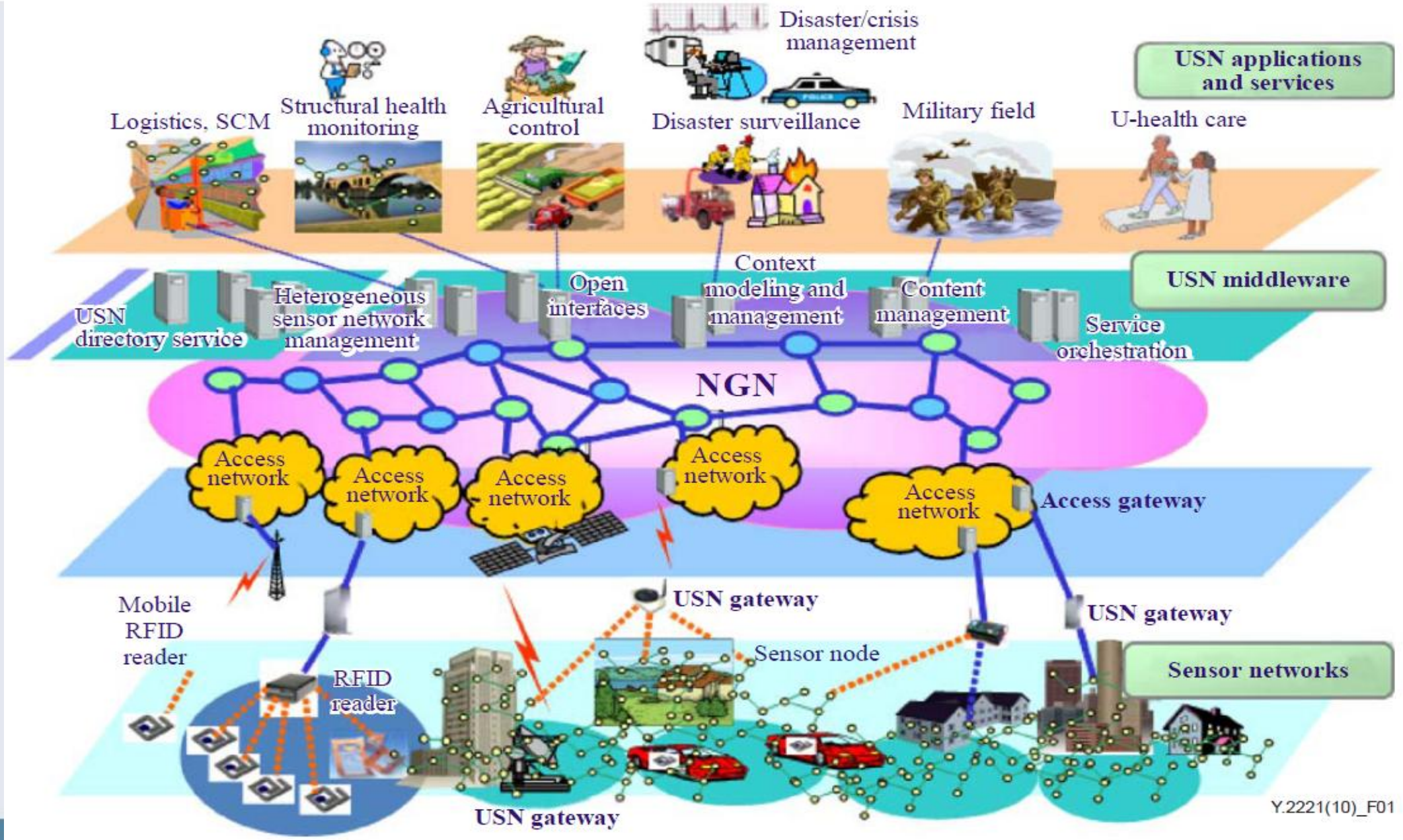
Yesterday in one picture



- Layers: change the radio, keep the application
- Wireshark shows every layer of every packet
- Devices dial out and keep the session open

Today: what travels on top - LoRaWAN, MQTT and dashboards

USN with related technical areas



Internet of Things World Forum (IoTWF) Reference Model

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Levels

7 **Collaboration & Processes**
(Involving People & Business Processes)

6 **Application**
(Reporting, Analytics, Control)

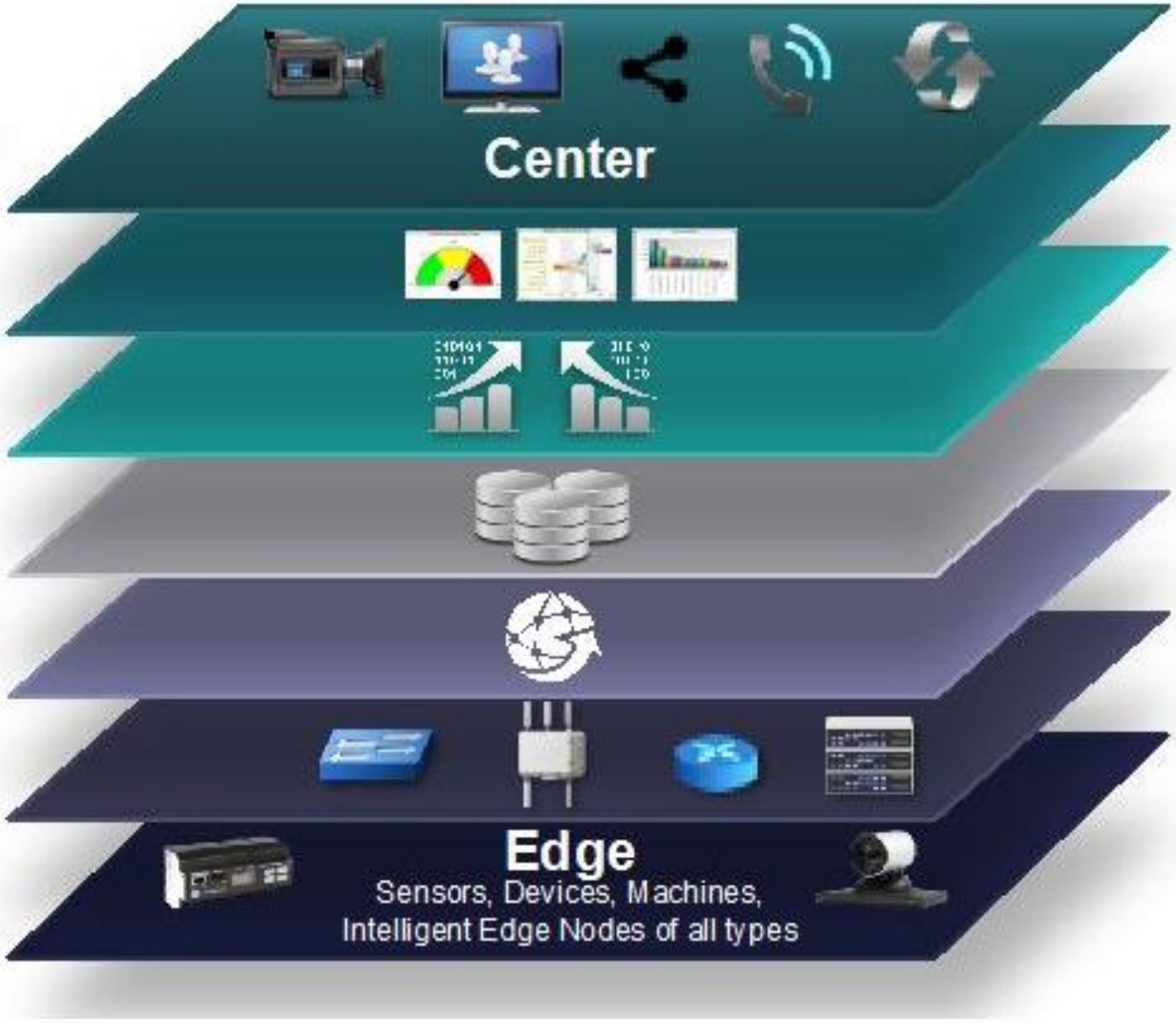
5 **Data Abstraction**
(Aggregation & Access)

4 **Data Accumulation**
(Storage)

3 **Edge Computing**
(Data Element Analysis & Transformation)

2 **Connectivity**
(Communication & Processing Units)

1 **Physical Devices & Controllers**
(The "Things" in IoE)



IoT: 4 layer Model

Integrated Applications



Information Processing







Network Infrastructure







Sensing and Identification

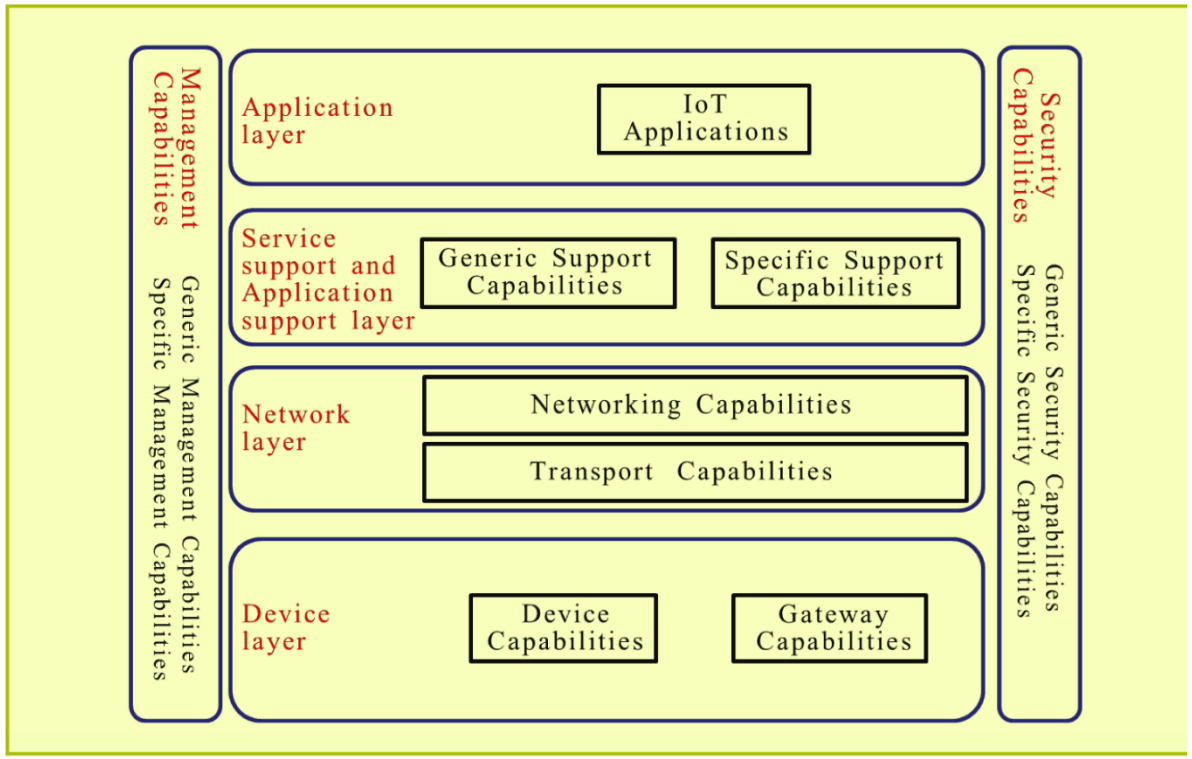






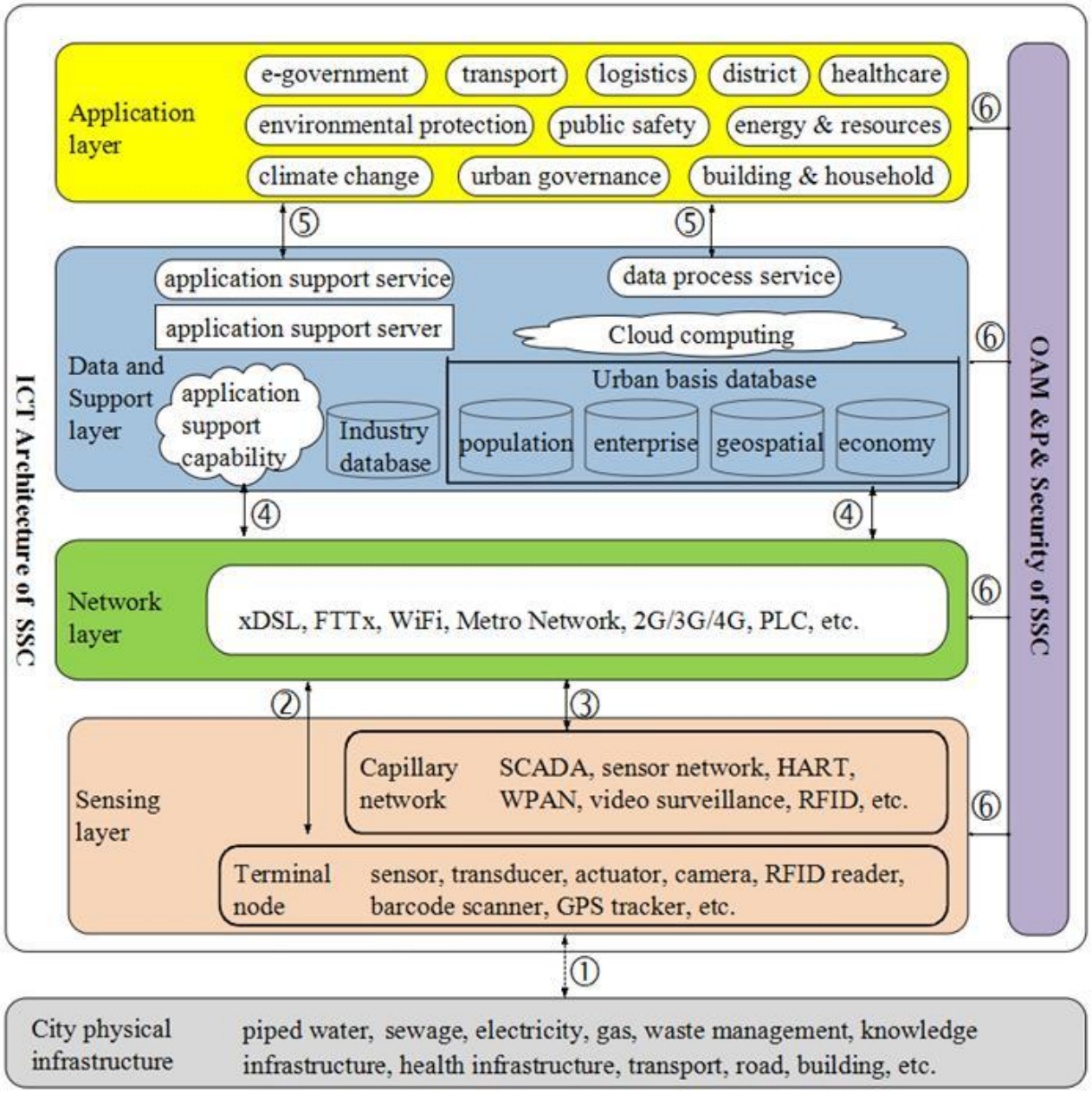


IoT reference model



Source: Recommendation **ITU-T Y.2060**

In IoT solutions supporting FC part of the application processing is executed directly at IoT objects and only when needed. More complex and resource-consuming tasks are transferred to higher level units (FC units) or directly to the cloud.



A multi-tier SSC (smart sustainable city) ICT architecture from communication view

Source: ITU-T Focus Group on Smart Sustainable Cities: *Overview of smart sustainable cities infrastructure*

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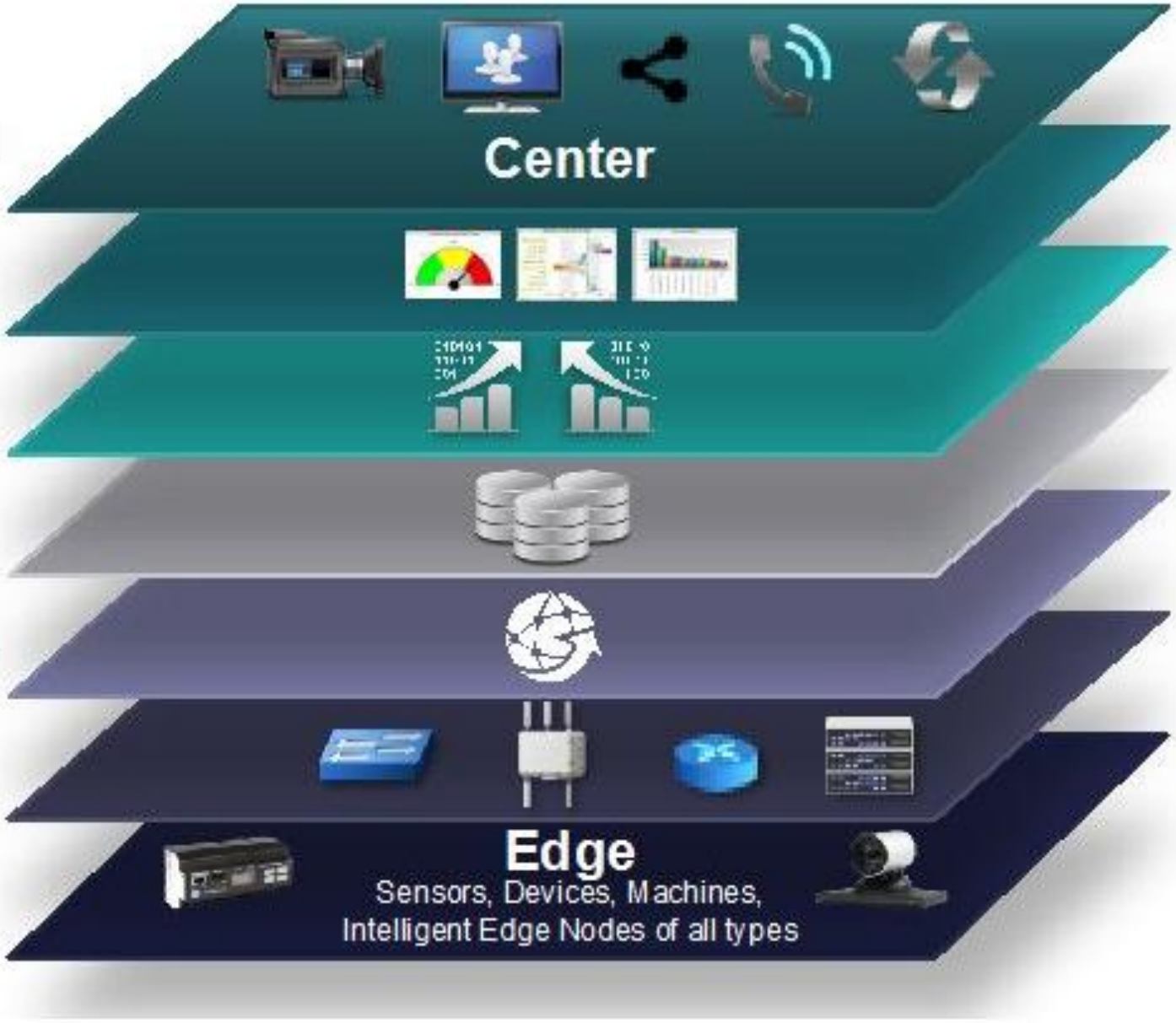
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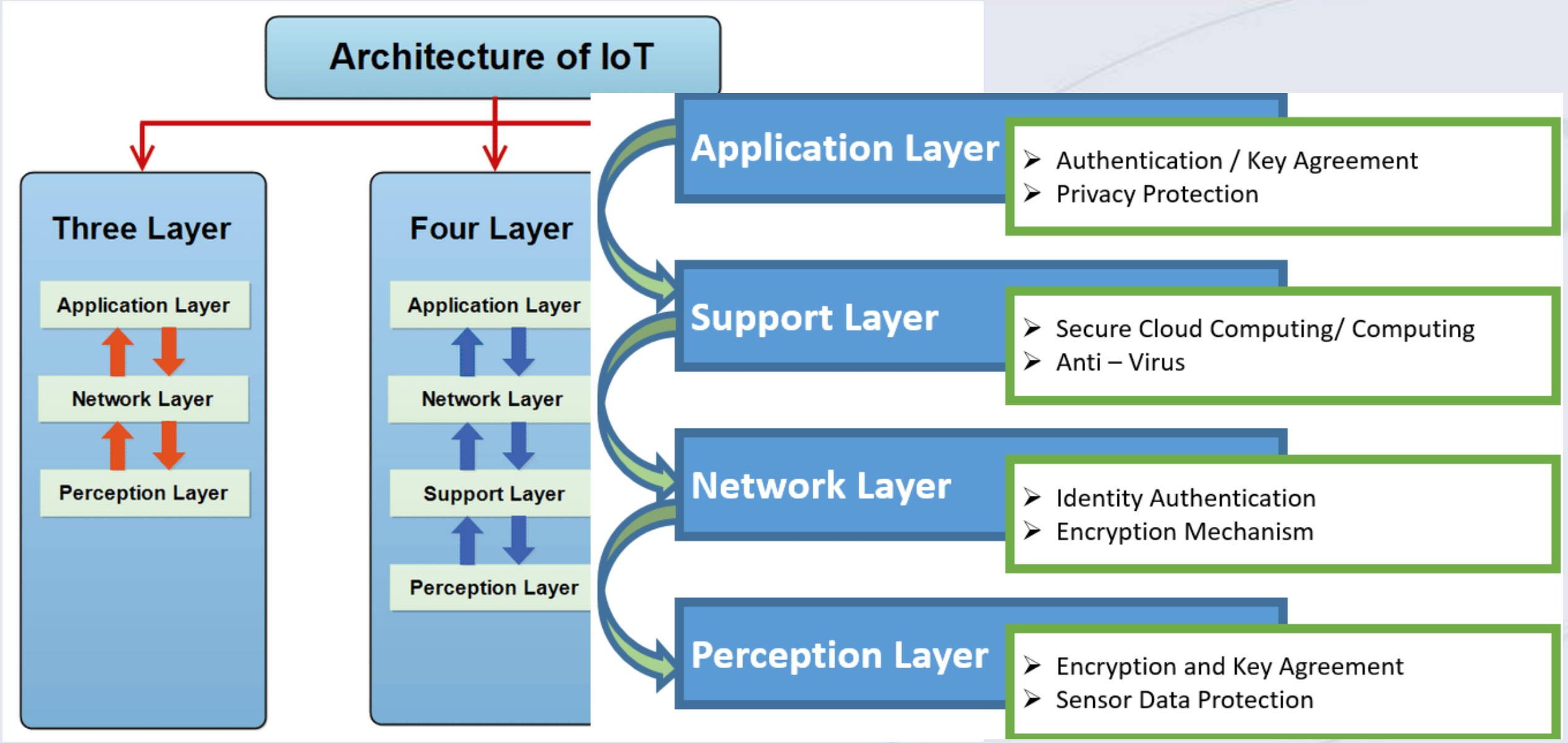
3 **Edge Computing**
(Data Element Analysis & Transformation)

2 **Connectivity**
(Communication & Processing Units)

1 **Physical Devices & Controllers**
(The "Things" in IoE)



Simplified Architecture



Edge, fog and cloud: where to decide

Edge - on the device

- Reacts in milliseconds
- Works with no network
- Tiny CPU and battery

siren when the door opens

Fog - the gateway

- Filters and combines data
- Keeps working if the link drops
- Translates radio to IP

buffers data during an outage

Cloud

- Unlimited storage and compute
- Fleet view, AI, dashboards
- Needs connectivity

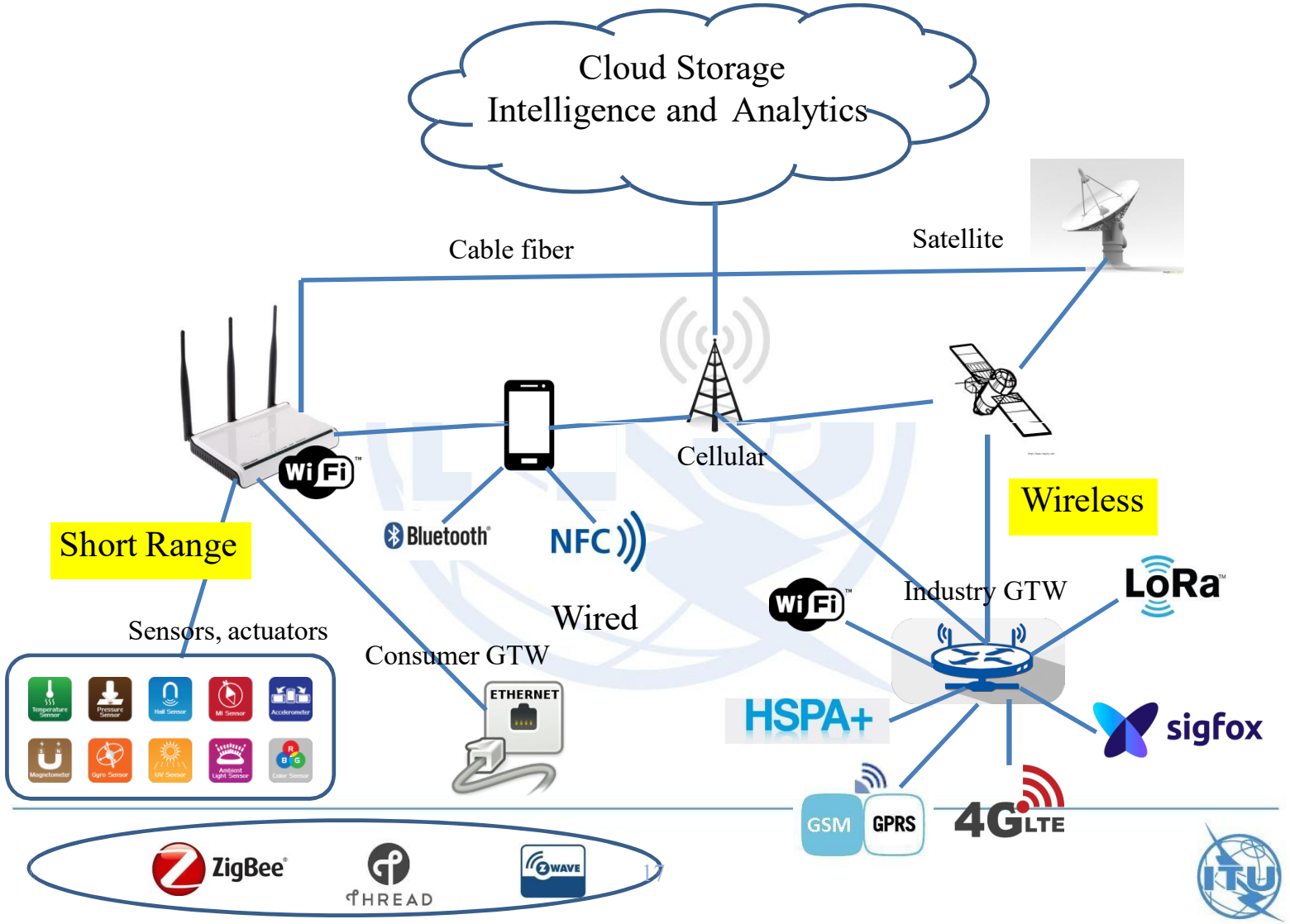
trends across 50 sites



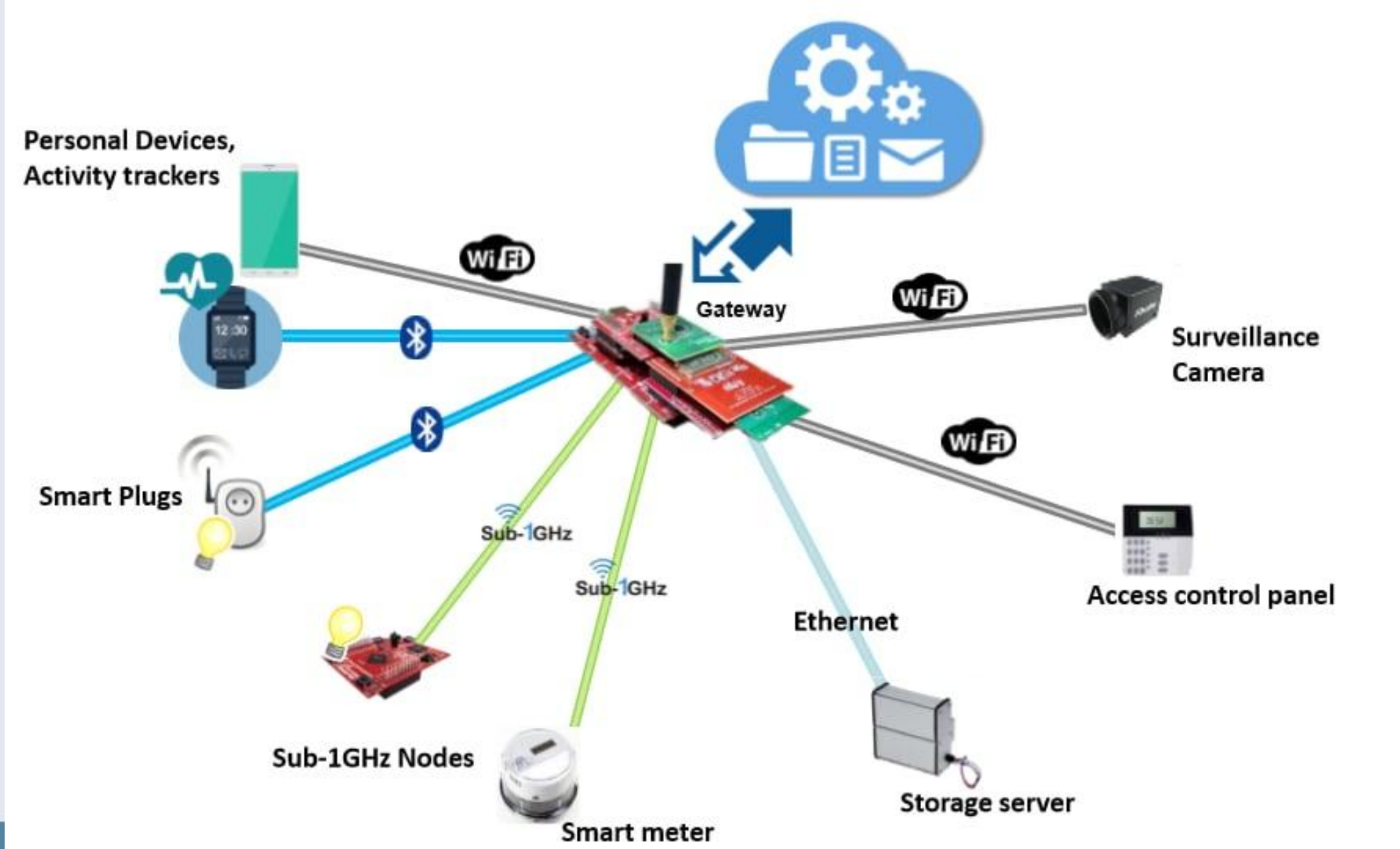
faster, works offline

more data, more context

IoT: the general architecture



IoT GateWays



Platforms come and go: plan for exit

When	AWS service	What happened
Jan 2025	IoT 1-Click	**shut down**
Jun 2025	Timestream LiveAnalytics	**no new customers**
Oct 2025	Fleet Hub	**shut down**
Dec 2025	IoT Analytics	**shut down**
Mar 2026	IoT FleetWise	**no new customers**
May 2026	IoT Events	**shut down**
Oct 2026	Greengrass V1	**support ends**

Lessons

Use open protocols: MQTT, LoRaWAN

Keep your data exportable

Wrap the vendor API in your own code

Earlier example: Google Cloud IoT Core (retired; its product page now redirects). Source: AWS General Reference - services in maintenance, sunset and full shutdown (fetched 7 Oct 2026)

Digital twin: a live copy of a thing



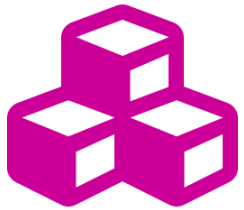
Physical site

pump, door, battery

measured state: temperature, door, level

ask it: what if? what next?

commands and settings



Digital twin

state + model + history

Smallest twin

AWS IoT Device Shadow: desired vs reported state - in our Day 4 lab

Bigger twins

3D model plus live data, e.g. AWS IoT TwinMaker (active, Oct 2026)

Standards

ISO/IEC JTC 1/SC 41 is the committee for IoT and digital twins

AIoT: intelligence moves to the edge

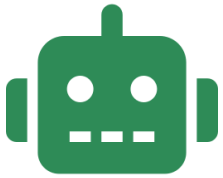
<1%

of IoT connections had real edge AI at the end of 2025

IoT Analytics, Feb 2026

Why move AI to the edge?

- Decide without the network
- Send results, not raw data
- Privacy: video never leaves the camera



Try it on a micro:bit: CreateAI

Record a few movements, train a model in the browser, then run it on the micro:bit with MakeCode - machine learning on a classroom board

Sources: IoT Analytics, State of enterprise IoT 2026 (4 Feb 2026); micro:bit CreateAI releases (v1.5.9, 23 Sep 2026)

LONG-RANGE CONNECTIVITY

LPWAN, cellular and satellite

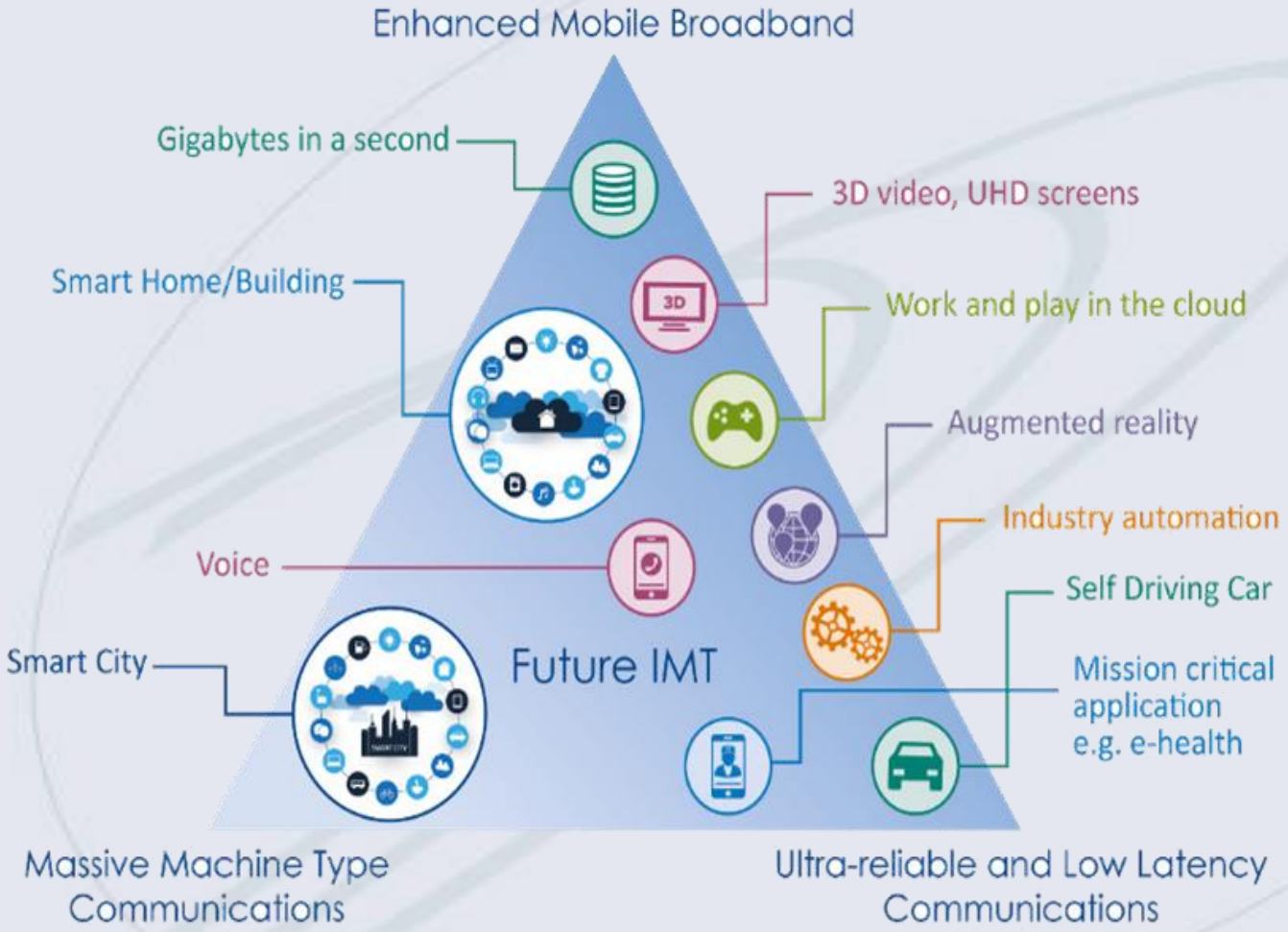
Long-range options at a glance

Technology	Spectrum	Who runs it	Speed	Typical use
LoRaWAN	unlicensed	you or an operator	very low	meters, sensors
Sigfox (0G)	unlicensed	UnaBiz + partners	tiny	alarms, trackers
NB-IoT	licensed	mobile operator	low	meters, fixed sensors
LTE-M	licensed	mobile operator	medium	trackers, wearables
LTE Cat-1 bis	licensed	mobile operator	medium	payment, telematics
5G RedCap	licensed	mobile operator	high	cameras, wearables
Satellite (NTN)	satellite	satellite operator	very low	remote assets

Licensed: an operator guarantees service, you pay per device. Unlicensed: free air, shared with everyone.

5G Motivation: The 3 Key 5G Use-cases

1. eMBB - faster&better throughput driven “wide pipe”
2. Massive MTC - cellular connectivity for IoT/IIoT
3. UR-LLC - assuring mission critical services functioning



Source: ITU-R, IMT 2020

Cellular IoT today

4.5 B
cellular IoT connections in 2025

7.8 B
forecast for 2031

+34%
Cat-1 bis module shipments, H1 2026

14
operators with 5G RedCap live

2G and 3G are going away

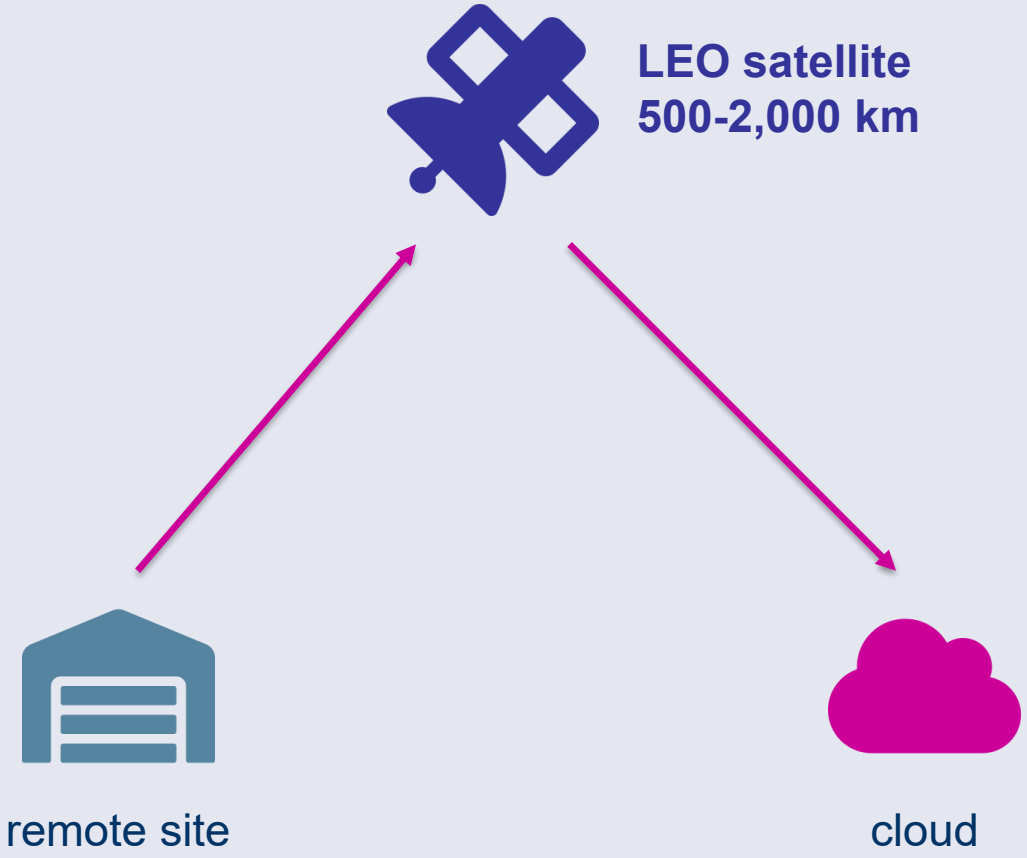
- By end 2025, 80 operators had shut 3G
- 46 operators had switched 2G off
- No 2G/3G modules for new sites

5G for IoT: RedCap

- Reduced-capability 5G (3GPP Rel-17)
- Chipsets to grow 82% a year to 2030
- Enhanced RedCap at scale from 2028

Sources: Ericsson Mobility Report, IoT connections outlook (2026); IoT Analytics, cellular IoT module shipments (16 Sep 2026) and State of enterprise IoT (4 Feb 2026); 3GPP RedCap page

Satellite IoT: coverage everywhere



- 3GPP Release 17: NB-IoT and LTE-M over satellite
- Release 19: store and forward when no ground station is in view
- Trade-offs: delay, small messages, cost per message
- Our site: satellite backhaul for the LoRaWAN gateway

Sources: 3GPP NTN overview (fetched 7 Oct 2026); IoT Analytics lists satellite integration as a 2026 IoT theme (Jan 2026)

LORA AND LORAWAN

Long range, low power, low data rate

LoRa is the radio, LoRaWAN the network

Your application
sensor data and commands

LoRaWAN
who may talk, keys, device classes

LoRa
chirp radio modulation

Radio band
Israel: 917-920 MHz (AS923-4)

LoRa - physical layer

- Chirp spread spectrum radio by Semtech
- Also used without LoRaWAN, e.g. Meshtastic

LoRaWAN - network protocol

- Open standard of the LoRa Alliance
- An ITU standard since 2021 (Y.4480)
- Specs: L2 1.0.4, Regional Param. 1.0.5

Day 1 again: layers let us change the radio without rewriting the application

LoRaWAN in 2026: the numbers

150 M
LoRaWAN devices deployed (Sep 2026)

~25%
yearly growth (CAGR)

340+
LoRa Alliance member organisations

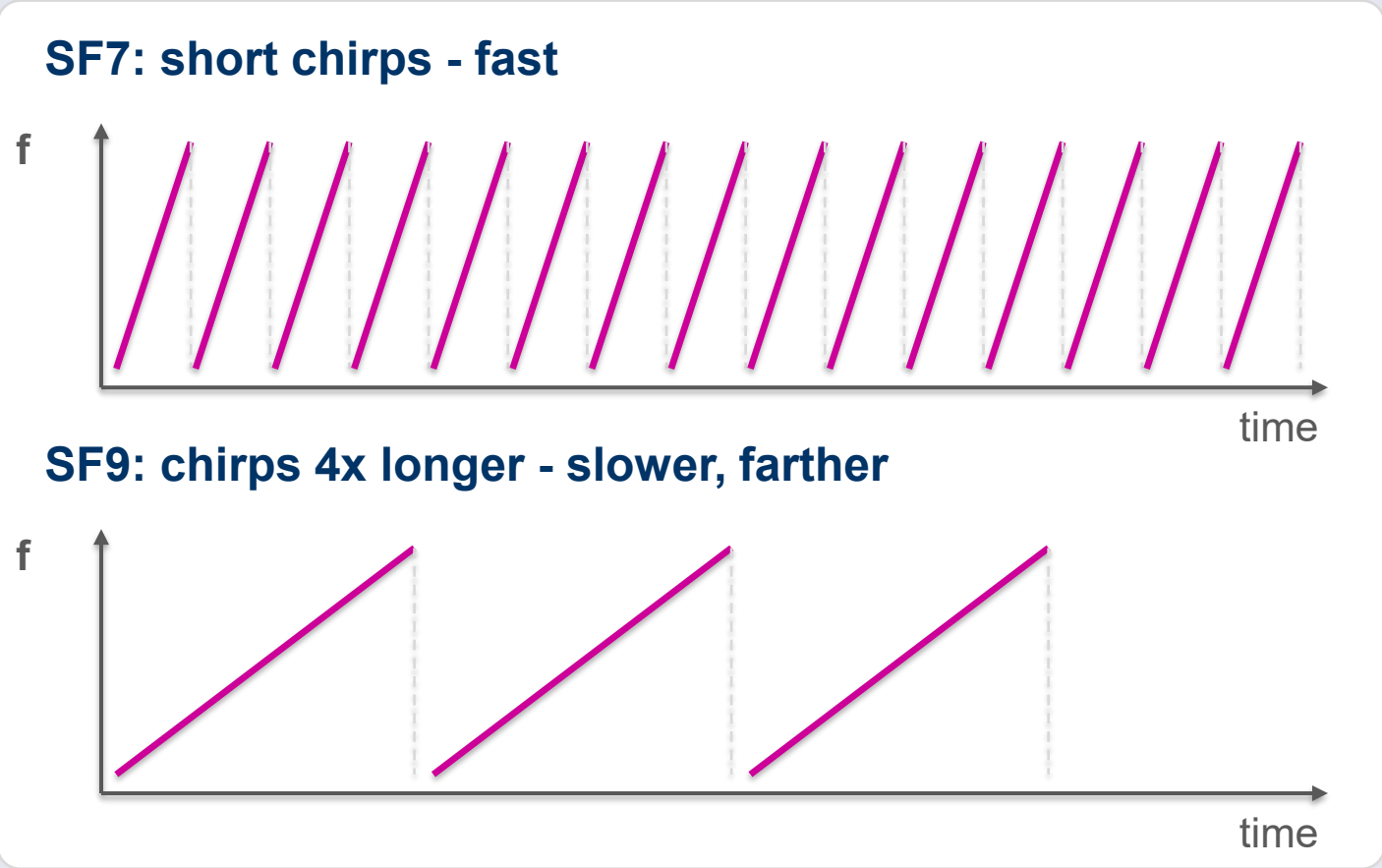
695+
certified device models

ZENNER Connect	Netmore	Veolia (France)	Arad (Israel)
11.6 M sensors 174,000+ gateways 15 countries	11.2 M active devices	4 M+ smart meters	1 M+ LoRa water meters and sensors, dozens of cities

The Alliance calls LoRaWAN the "fourth pillar" next to cellular, Wi-Fi and Bluetooth

Sources: LoRa Alliance press release, 22 Sep 2026 (operator figures are self-reported); Arad Group, 16 Nov 2025

Chirps: how LoRa survives the noise



- Each symbol is a sweep across the 125 kHz channel
- The receiver finds the sweep even below the noise
- Longer chirps carry the same bits more slowly - and farther

Like speaking slowly in a noisy room: understood from farther away, but it takes longer

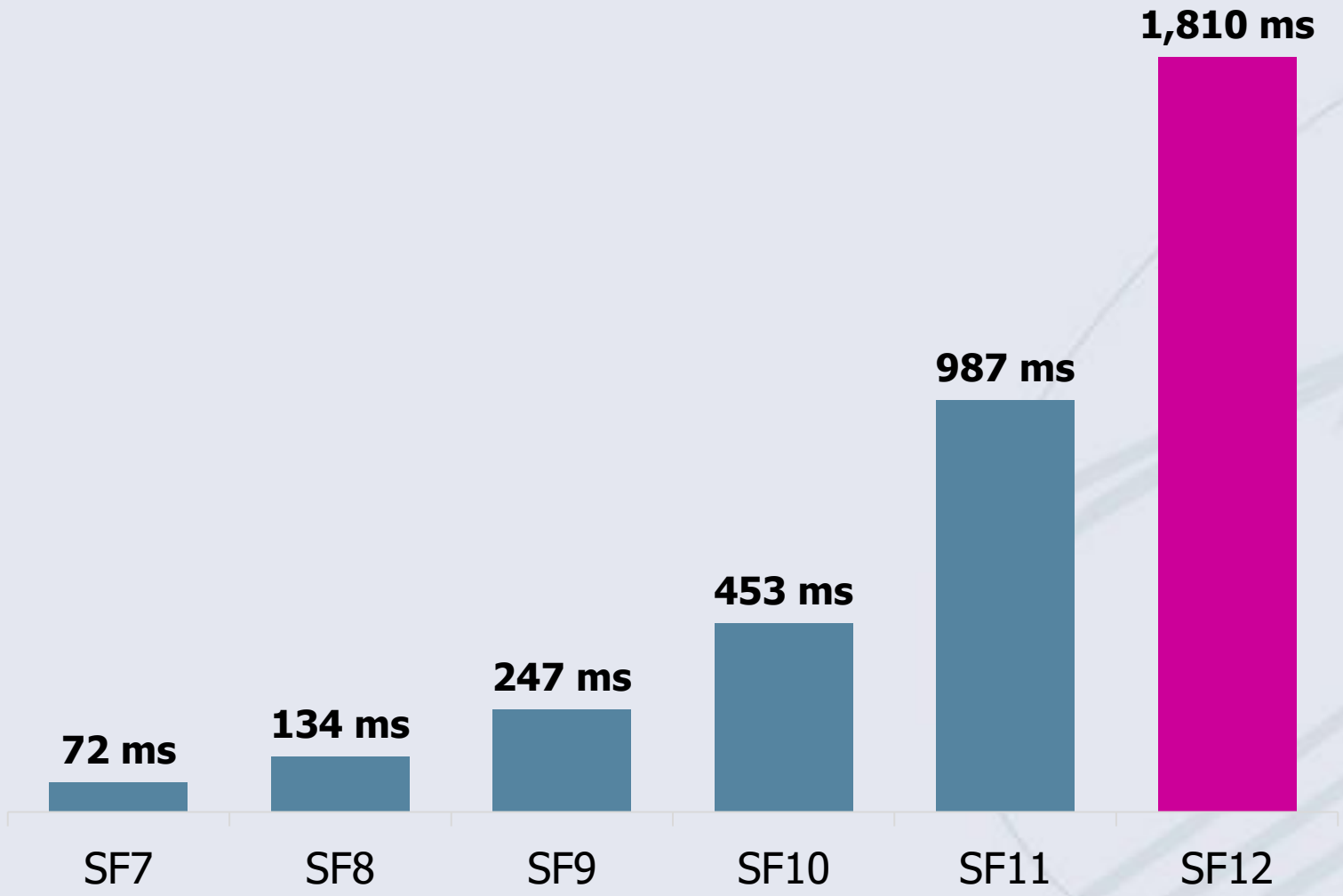
Spreading factor: range against speed

Spreading factor	Bit rate	Airtime, 20-byte message	Sensitivity	Range (theory)
SF7	5,470 bit/s	72 ms	-123 dBm	1.0x
SF8	3,125 bit/s	134 ms	-126 dBm	1.4x
SF9	1,760 bit/s	247 ms	-129 dBm	2.0x
SF10	980 bit/s	453 ms	-132 dBm	2.8x
SF11	440 bit/s	987 ms	-134.5 dBm	3.8x
SF12	250 bit/s	1,810 ms	-137 dBm	5.0x

Each step up: half the speed, double the airtime, about 1.4x the range

Bit rates: LoRa Alliance RP002 (125 kHz). Airtime: Semtech formula, 20 B + 13 B LoRaWAN overhead, CR 4/5. Sensitivity: TTN docs. Range: free-space theory.

Airtime doubles with every SF step



25x

more airtime at SF12 than at SF7, same message

- Airtime costs battery
- Airtime blocks the channel for others
- Use the fastest SF that still arrives

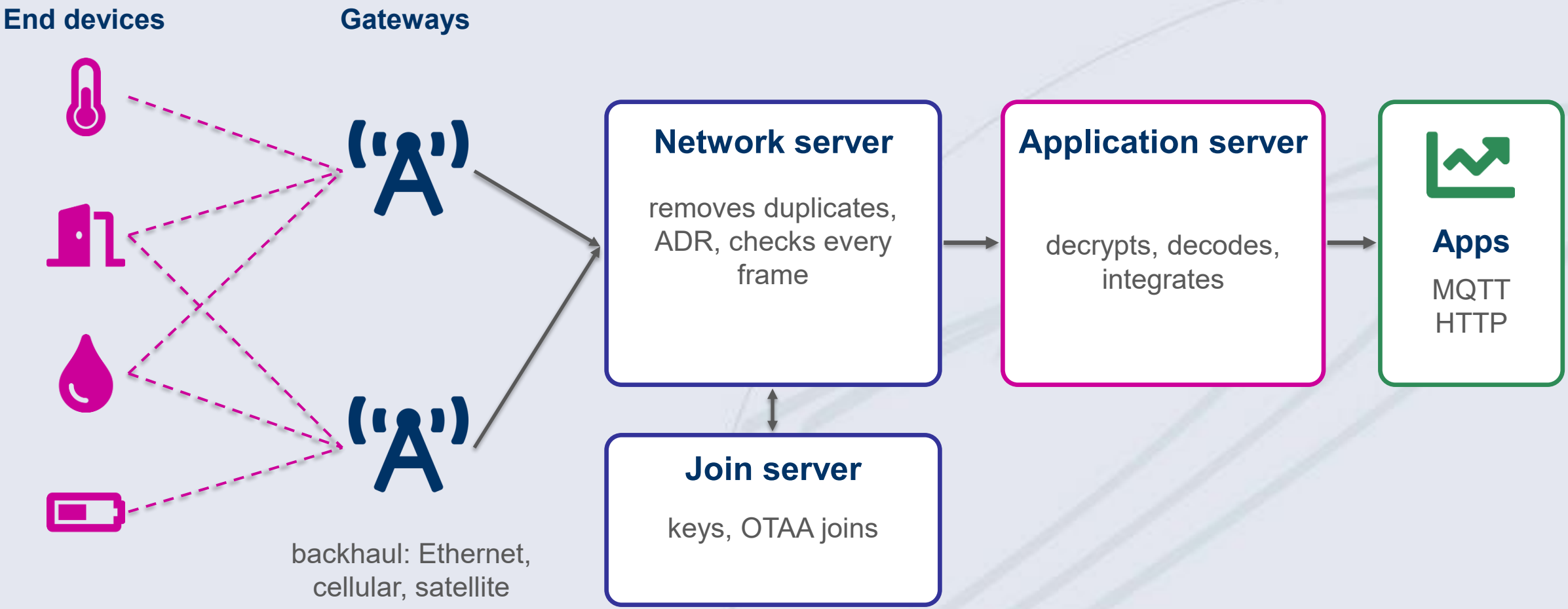
Link budget: why LoRa reaches so far

Transmit power		Receiver sensitivity		Link budget
+16 dBm	-	**-137 dBm**	=	**153 dB**
EU868 limit (example)		SF12, 125 kHz		what the signal may lose

10 km of free space at 917 MHz costs about 112 dB: 41 dB left for walls, trees, hills

- +6 dB of budget = double the distance (free space)
- SF12 instead of SF7: +14 dB = up to 5x the distance
- Height beats power: put the gateway high
- Typical: a few km in town, 15+ km in open country

LoRaWAN architecture: a star of stars



Gateways forward everything

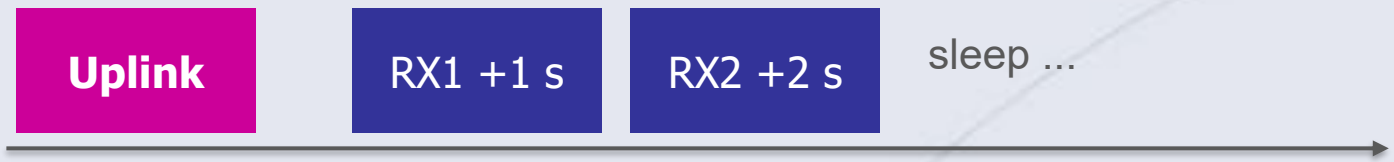
No pairing to one gateway

The server keeps one copy

Device classes: A, B and C

Class A

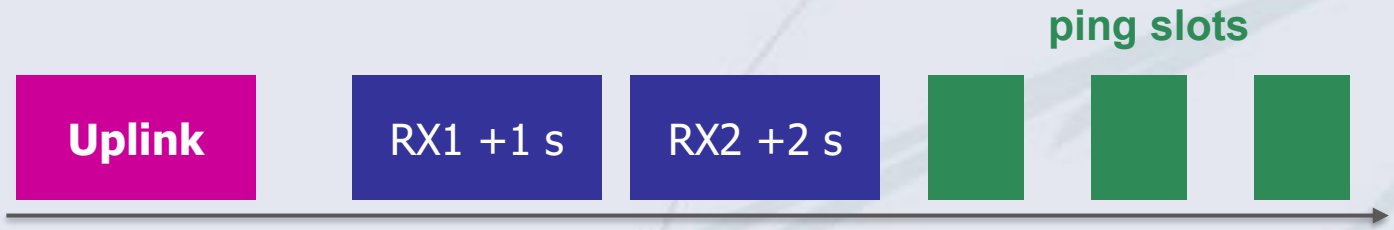
every device



Battery: best
Downlink waits for the next uplink

Class B

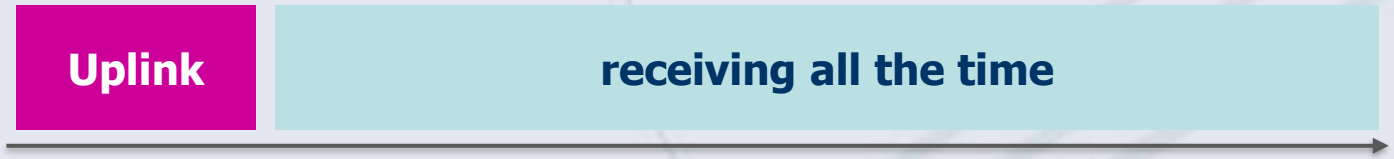
scheduled



Battery: good
Downlink at the next ping slot

Class C

always listening



Battery: mains power
Downlink at once

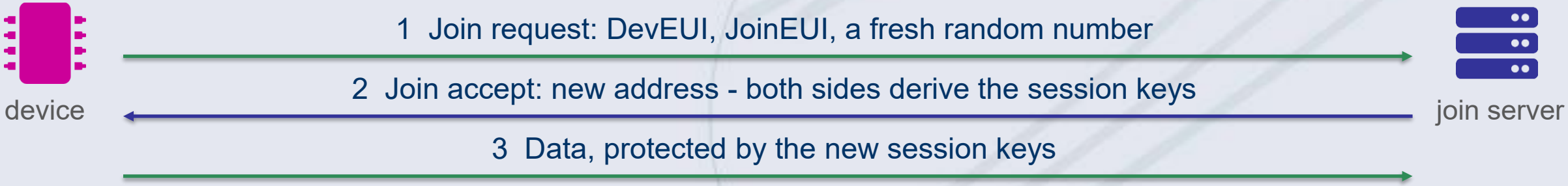
Joining the network: OTAA versus ABP

OTAA - over-the-air activation

- Stores DevEUI, JoinEUI and AppKey
- New session keys at every join
- Can move to another network

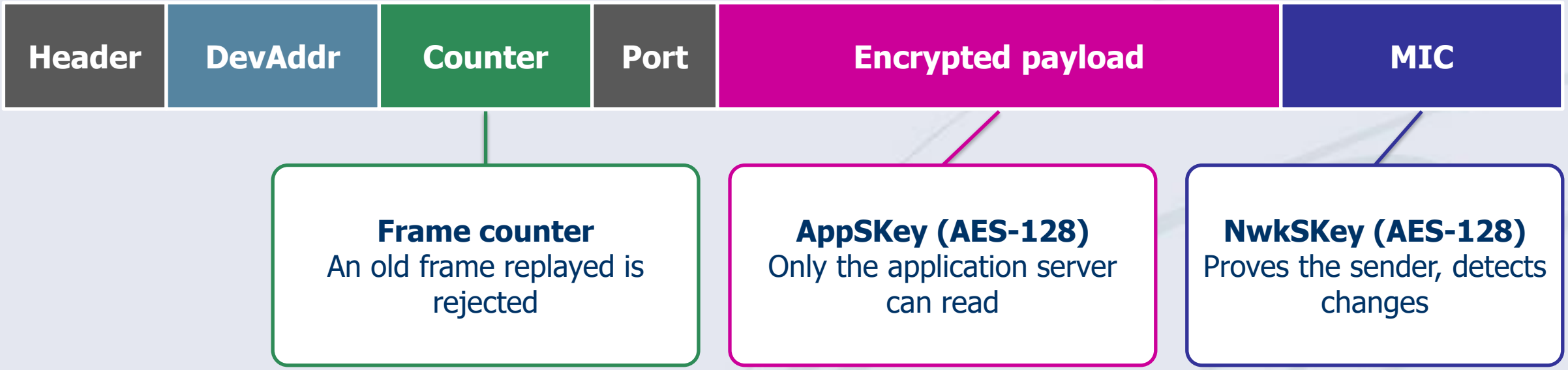
ABP - set at the factory

- Address and keys burned in, no join
- Same keys forever
- Counter reset after reboot = replay risk



Rule: use OTAA. Keep ABP for lab tests only.

LoRaWAN security: two keys, two jobs



- Gateways and the network operator cannot read your data
- LoRaWAN 1.1 splits the network key in three for tighter control
- Keys must be unique per device - never one AppKey for all

Adaptive data rate (ADR)



Near	Middle	Far
SF7 - 72 ms low transmit power	SF9 - 247 ms	SF12 - 1,810 ms full power

distance from the gateway

1. The network server watches the signal quality of recent uplinks
2. If there is margin: switch to a faster SF and less power
3. Saves battery and airtime - but not for moving devices

Payloads are bytes, not JSON

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JSON - easy to read

```
{"temperature":23.45,
 "battery":3.1,"door":true}
```

47 bytes

Packed - easy to send

```
09 29 1F 01
```

09 29 = temperature x 100
1F = battery x 10
01 = door open

4 bytes

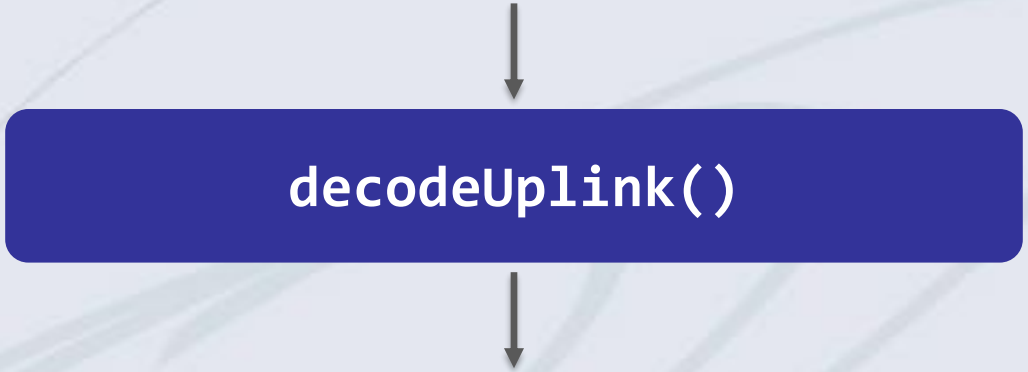
At SF10: JSON 698 ms on air, packed 330 ms

Max payload (EU868 example): 51 bytes at SF10-SF12, 242 at SF7-SF8

The decoder turns bytes into JSON

```
function decodeUplink(input) {
  var b = input.bytes;
  return {
    data: {
      temperature: ((b[0] << 8) | b[1]) / 100,
      battery: b[2] / 10,
      door: (b[3] & 1) === 1
    }
  };
}
```

```
09 29 1F 01
```




```
{"temperature": 23.45,
 "battery": 3.1,
 "door": true}
```

ChirpStack and The Things Stack both run a JavaScript decodeUplink() per device type

Airtime rules: duty cycle and fair use

EU868: 1% duty cycle

- 36 s of airtime per hour
- SF7: up to 500 messages/hour
- SF12: up to 19 messages/hour

TTN community fair use

- 30 s of uplink airtime per device per day
- 10 downlinks per day
- SF7 ~417/day, SF12 ~16/day

Israel: AS923-4

- Limits are set by the Ministry of Communications
- Check them before you deploy
- Private networks set their own fair use

Send less, send smaller, use the fastest SF that works

EU duty cycle and TTN fair use: TTN docs (fetched Oct 2026). Message counts: 20-byte message, computed airtime.

Design check: will our site fit?

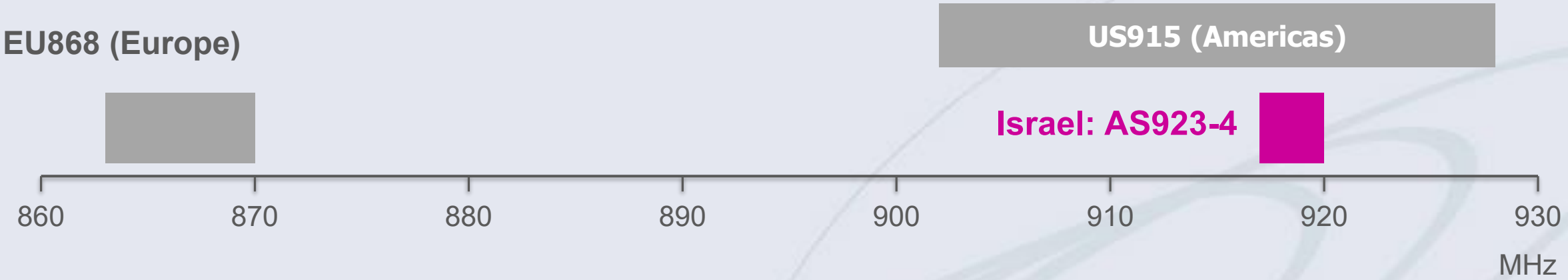
Each sensor: one 20-byte message every 10 minutes = 144 messages a day

Spreading factor	Airtime per message	Airtime per day	Fits 30 s a day?
SF7	72 ms	10 s	**yes**
SF9	247 ms	36 s	**no**
SF12	1,810 ms	261 s	**no**

If it does not fit:

- Raise the gateway: a faster SF becomes possible
- Send every 15 minutes, or only when something changes
- Pack the bytes (previous slides)

Israel: AS923-4, not EU868



- Band 917-920 MHz
- Default channels 917.3 and 917.5 MHz
- Reply window RX2: 917.3 MHz

- ChirpStack region file: as923_4
- TTN plan IL_917_920_TTN (8 channels)
- EU433 and AS923-3 dropped for Israel (2019)

Buy devices and gateways that support AS923-4. Check the MoC power limits.

Who runs the network server?

ChirpStack 4.19 - open source

- Run it yourself: a private network
- Needs an MQTT broker, PostgreSQL, Redis

The Things Stack 3.36

- Open source and a commercial cloud
- Community network: The Things Network

AWS IoT Core for LoRaWAN

- Managed network server inside AWS
- Active, no end-of-life notice (Oct 2026)

Network operators

- You buy coverage, not servers
- e.g. ZENNER, Netmore; Arad in Israel

Public network: someone else's gateways. Private: your gateways, your data.

An uplink, as your application sees it

```
topic: application/7d0c.../device/0004a30b001c2f3a/event/up
```


```
{
  "deviceInfo": {
    "applicationName": "remote-site",
    "deviceName": "pump-room-01",
    "devEui": "0004a30b001c2f3a"
  },
  "fCnt": 1287,
  "fPort": 1,
  "data": "CSkfAQ==",
  "object": {"temperature": 23.45,
             "battery": 3.1, "door": true},
  "rxInfo": [{"gatewayId": "a840411f2b3c4d5e",
              "rssi": -112, "snr": 4.5}],
  "txInfo": {"frequency": 917300000,
    "modulation": {"lora": {"spreadingFactor": 9}}}
}
```


object - decoded by your decoder

rssi / snr - signal quality at each gateway

frequency - 917.3 MHz, the Israeli band

fCnt - the frame counter (anti-replay)

Beyond classic LoRaWAN

Meshtastic: LoRa mesh

- Open-source text and GPS messaging
- Phones connect over Bluetooth
- Backup comms when networks fail

Relays

- A relay device reaches basements and tunnels
- Spec TS011 1.0.1, guide TR016 (Aug 2026)

LR-FHSS

- Frequency hopping instead of one channel
- Very slow, very robust: 162 or 325 bit/s

Easier onboarding (2026)

- Scan a QR code to add a device (TS018)
- Network fetches the device profile (TS014)

Sources: LoRa Alliance specification pages (TS011 2024, TR016 Aug 2026, TS014/TS018 Jul 2026); RP002 data-rate table; Meshtastic firmware releases (2026)

Attacking and defending LoRaWAN

Threat	Defence
Jamming the radio	Alarm on missing data and odd signal levels
ABP counter reset -> replay	Use OTAA; never reset frame counters
One AppKey for all devices	Unique keys per device, kept in a join server
Gateway link in clear text (UDP)	Basics Station over TLS, or a VPN
Stolen device -> keys extracted	Secure element, tamper alarm, revoke the device
Fake gateway or server	Authenticate both ends of the backhaul

Security must cover the device, the radio, the backhaul and the cloud

MESH NETWORKS

Zigbee, Thread and Matter - IoT inside buildings

IEEE 802.15.4: the radio under the mesh

2.4 GHz

worldwide band

16 channels

numbered 11 to 26

250 kbit/s

data rate

127 bytes

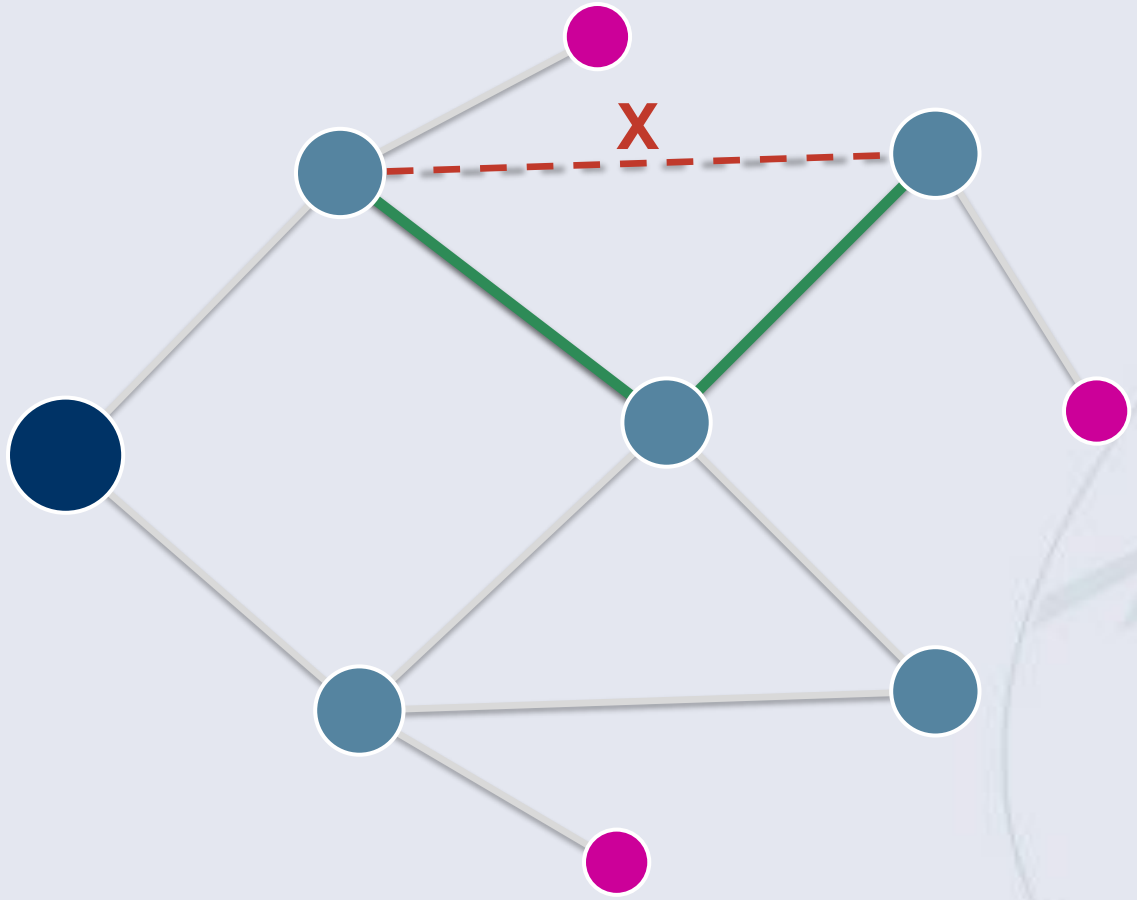
largest frame

Who uses it

- Zigbee: lights, sensors, meters
- Thread: the IPv6 mesh under Matter
- WirelessHART, ISA100: factories
- New: sub-GHz versions (Zigbee 'Suzi', Thread work started 2026)

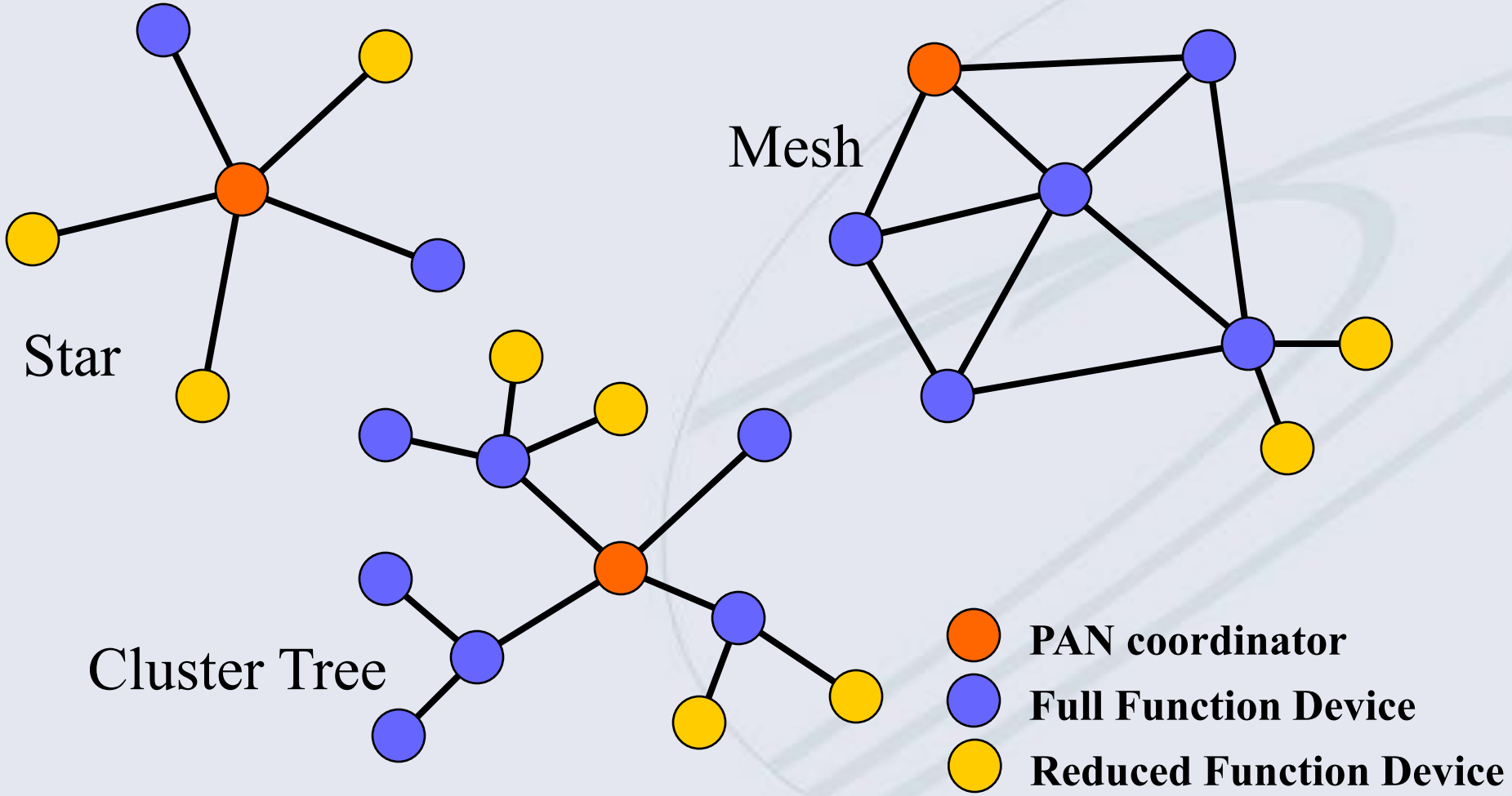
Low power and short range - the mesh provides the reach

Mesh basics: many paths, no weak link



- Routers relay for others: more devices = more coverage
- Battery devices sleep and do not relay
- A broken link heals itself **through another route**
- Same idea as LoRa's 'many gateways' - but every node helps

Topology Models



Zigbee, Thread and Matter in 2026

	Zigbee	Thread	Matter
What it is	full stack with its own network layer	IPv6 mesh network	application layer
Runs on	802.15.4 - 2.4 GHz, now also sub-GHz	802.15.4	Wi-Fi, Thread, Ethernet
Current version	Zigbee 4.0 (Nov 2025)	Thread 1.4 (spec 1.4.1)	Matter 1.6 (Jun 2026)
What is new	Suzi sub-GHz: certification since Sep 2026	shared credentials across vendors	NFC setup, Joint Fabric
Owner	Connectivity Standards Alliance	Thread Group	Connectivity Standards Alliance

Matter is one language for smart devices; Thread is one of its networks

Matter in one picture

Matter: one data model for lights, locks, sensors, cameras

IPv6

Wi-Fi | **Thread 802.15.4 mesh** | **Ethernet**



a Thread border router joins Thread to the home network

4,600+ distinct certified Matter products in the public registry (Oct 2026)

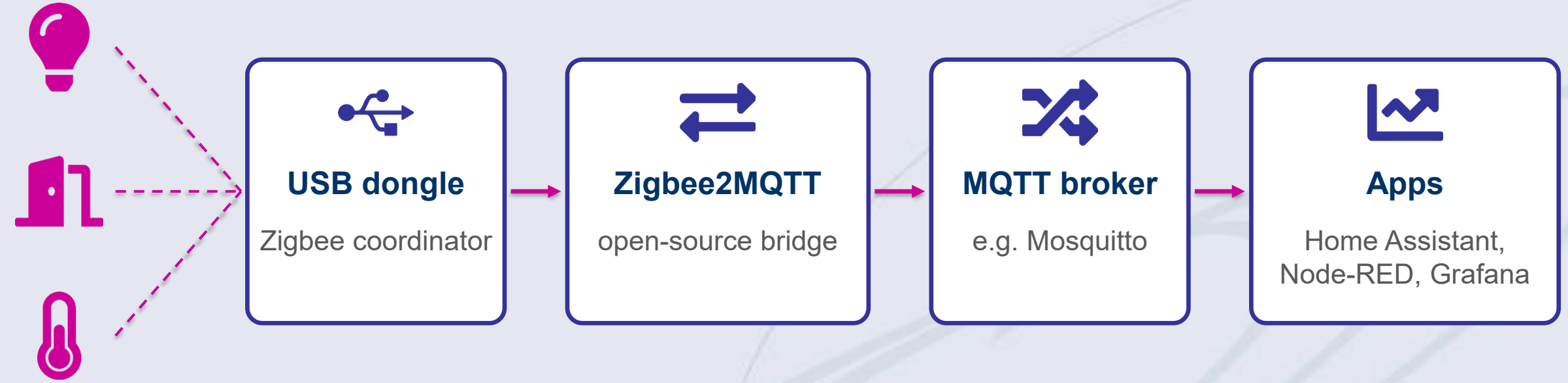
Setup

- Scan the QR code
- Over BLE, Wi-Fi only or NFC (1.6)

Many apps, one device

- Several ecosystems control it
- Joint Fabric (1.6): shared admin

Zigbee2MQTT: from mesh to MQTT




```
zigbee2mqtt/hall_door        {"contact": false, "battery": 97}
zigbee2mqtt/hall_light/set   {"state": "ON"}
```

Supports 5,798 devices from 608 vendors (v2.14.2, Oct 2026). MQTT is the meeting point - next section.

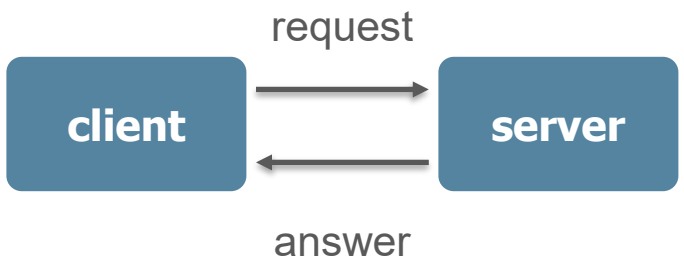
MESSAGING AND MQTT

How IoT data moves: publish / subscribe

Three ways to move a message

Request - response

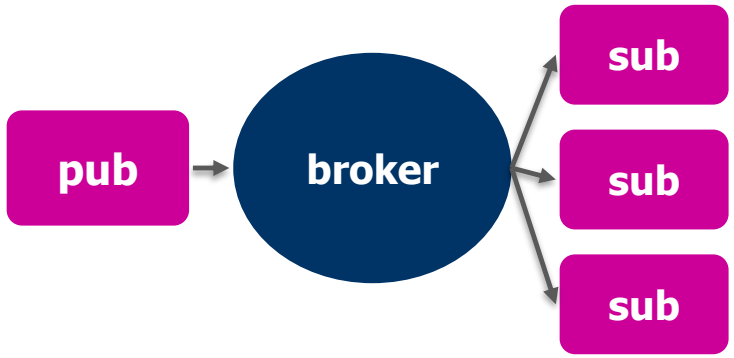
e.g. HTTP



- Ask and wait for the answer
- Both sides must be up

Publish - subscribe

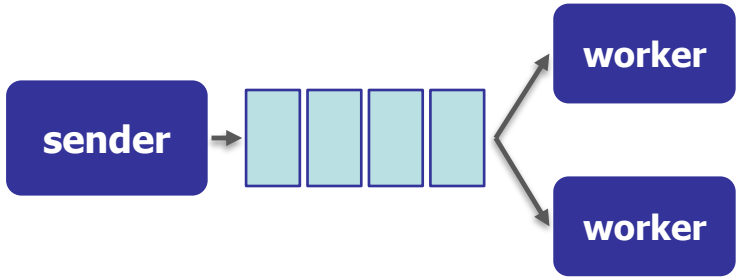
e.g. MQTT



- Senders and receivers never meet
- One message, many receivers

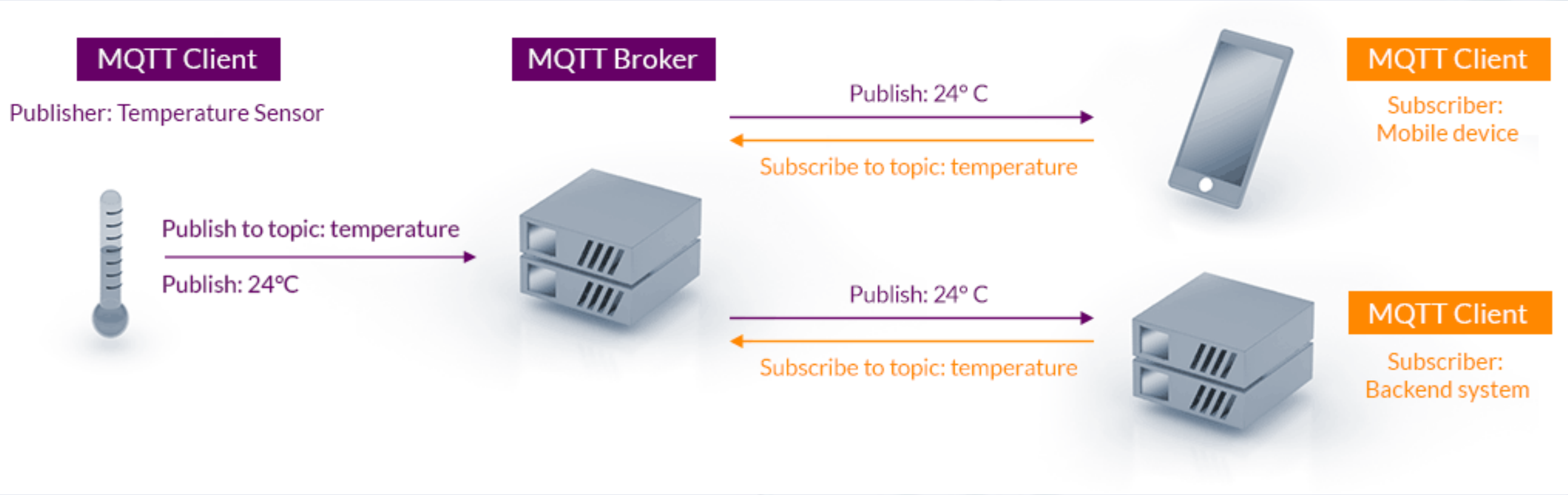
Queue

e.g. AMQP, Kafka, SQS



- Each message handled once
- Absorbs peaks of traffic

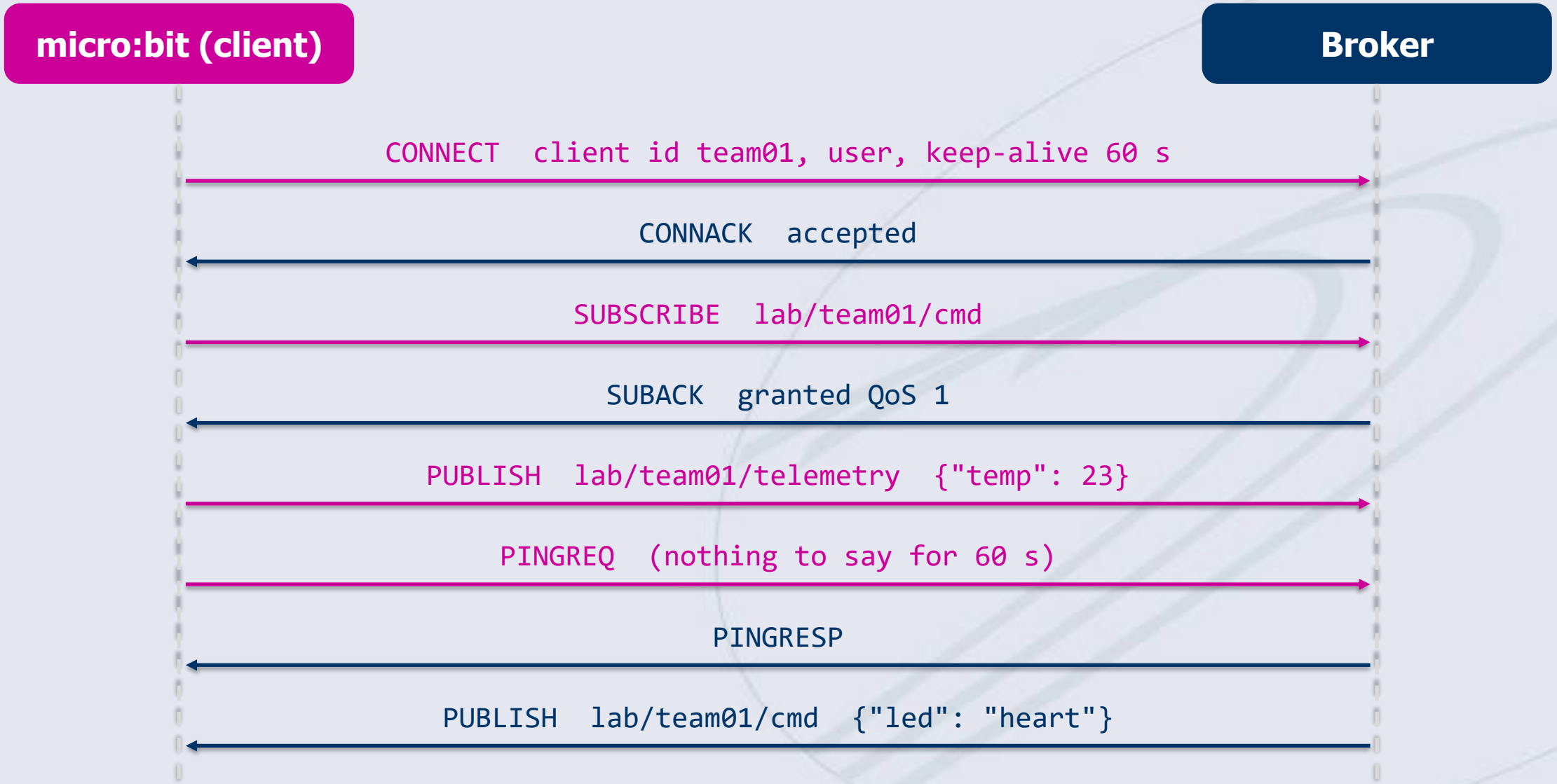
MQTT Publish / Subscribe Architecture



MQTT in five facts

1.  Clients always dial out to the broker - works behind NAT
2.  TCP port 1883, or 8883 with TLS (also over WebSockets)
3.  Tiny: the fixed header is just 2 bytes
4.  Standards: 3.1.1 (ISO/IEC 20922) and 5.0 (OASIS, 2019)
5.  Spoken by AWS IoT Core, ChirpStack, The Things Stack, Zigbee2MQTT

A session, step by step



Topics: the address of a message

```
site/tel-aviv/pump-1/temperature
```

site location device measurement

- Levels are separated by /
- Case-sensitive: Temp is not temp
- No leading /, no spaces
- $ topics are reserved ($SYS, $aws)
- IDs in the topic, values in the payload

Day 4 lab plan

- lab/team01/telemetry
- lab/team01/cmd
- each team only its own

Wildcards: + one level, # all below

Subscription	Receives	Does not receive
`site/+/pump-1/temperature`	site/haifa/pump-1/temperature	site/haifa/pump-2/temperature
`site/tel-aviv/#`	site/tel-aviv/door site/tel-aviv/pump-1/battery	site/haifa/door
`+/+/+/battery`	site/eilat/pump-3/battery	site/eilat/battery
`#`	everything - debugging only!	-

Wildcards work only when subscribing - you always publish to one exact topic

Quality of service: 0, 1 and 2

QoS 0 - at most once



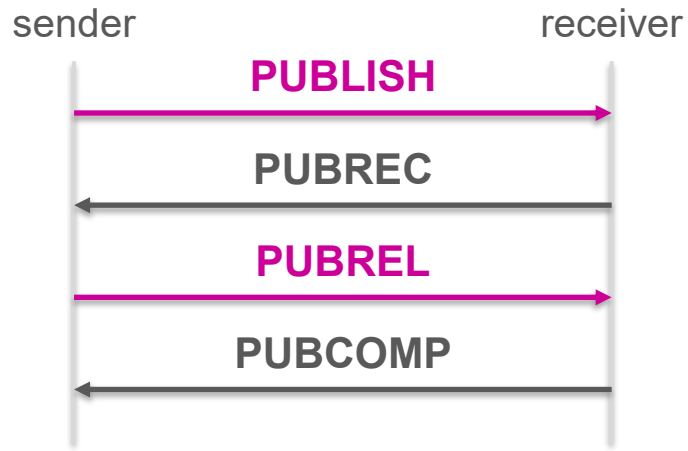
Fire and forget. May be lost.

QoS 1 - at least once



Repeated until acknowledged.
May arrive twice.

QoS 2 - exactly once



Four packets. Slowest.

not on AWS IoT Core

Telemetry: QoS 0. Alarms and commands: QoS 1 - and ignore duplicates.

Keep-alive and the last will



- Keep-alive: the client promises to say something every N seconds
- Silent for 1.5 x N: the broker declares the client gone
- Last will: a message the broker sends on the client's behalf
- Our site: 'site offline' without any extra code on the device

Retained messages: the last known value



- The broker keeps one last message per topic
- New subscribers get it immediately - no waiting for the next reading
- Perfect for status and settings, not for every measurement
- Delete it by publishing an empty retained message

AWS IoT Core: wildcard subscriptions do not receive retained messages

Securing MQTT

LOGTEL
It's more than knowledge



Encrypt TLS on port 8883 - never plain 1883 across the Internet



Authenticate user + password, or a client certificate (AWS IoT: X.509)



Authorise per-topic rules: team01 may only use lab/team01/...



No anonymous Mosquitto 2.x refuses anonymous remote clients by default

An open broker on the Internet = anyone can read your data and send commands

Brokers: which one?

Broker	Type	Version (Oct 2026)	Notes
Mosquitto	open source, one server	2.1.2 (Feb 2026)	light - labs and gateways
EMQX	clustered, enterprise	6.3.1 LTS (Sep 2026)	very large fleets; check the licence
AWS IoT Core	managed cloud service	always current	QoS 0/1, X.509 - our Day 4 lab
The Things Stack	built into the LoRaWAN server	3.36 (Aug 2026)	MQTT 3.1.1, QoS 0 only

Same protocol everywhere: a client written for Mosquitto talks to AWS IoT Core

Mosquitto quick start

LOGTEL
It's more than knowledge

mosquitto.conf

```
listener 1883
allow_anonymous false
password_file passwd
```

- Without a listener line: localhost only
- Without a password file: nobody gets in

Commands

```
mosquitto_passwd -c passwd team01
mosquitto -c mosquitto.conf -v
mosquitto_sub -h localhost -u team01 -P <password> -t "lab/#" -v
mosquitto_pub -h localhost -u team01 -P <password> -t lab/team01/telemetry -m 23.5
```

Mosquitto 2.1.2 (Feb 2026) - Windows, Linux, macOS and Raspberry Pi

Tools to see MQTT

MQTT Explorer

- Topic tree with live values
- Last full release 2019 - fine for demos

MQTTX 1.13

- Desktop app and command line
- Released Sep 2026

mosquitto_sub / pub

- Command line, ships with Mosquitto
- Quick tests and scripts

AWS IoT MQTT test client

- In the AWS console
- Subscribe to # - used on Day 4

MQTT in Wireshark

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Port 1883 - no TLS

```
MQ Telemetry Transport Protocol
  Connect Command
    User Name: team01
    Password: s3cret!
  Publish Message [lab/team01/telemetry]
    Message: 23.5
```

Anyone on the path reads everything

Port 8883 - TLS

```
Transport Layer Security
  TLSv1.3 Record Layer:
    Application Data
    Encrypted Application Data:
      8f3a91c4e07b...
```

Only sizes and timing are visible

Filters to try:

```
mqtt    tcp.port == 1883    mqtt.topic contains "team01"
```

MQTT and its neighbours

Protocol	Runs over	Best for
MQTT	TCP	devices to cloud, many-to-many
MQTT-SN 1.2	UDP, radio links	tiny sensors without TCP (2.0 in draft)
CoAP	UDP	web-style requests for tiny devices
Sparkplug 3.0	on top of MQTT	industrial topic and payload rules
AMQP	TCP	business messaging and queues
Kafka	TCP	data-centre streaming backbone
HTTP	TCP	simple requests, webhooks

Our topic plan for Day 4

```
lab/team01/telemetry           micro:bit -> cloud   {"temp": 23, "light": 120}
lab/team01/cmd                 cloud -> micro:bit   {"led": "heart"}
$aws/things/team01/shadow/...  device state: desired / reported
```

One identity per team

password at the gateway, certificate in AWS

One policy for all

each team may only use its own topics

QoS 0 or 1

AWS IoT Core has no QoS 2

Day 4: these topics in the AWS console and in Grafana - in Wireshark only where MQTT runs without TLS

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TIME SERIES AND GRAFANA

Storing and showing what the sensors say

Time-series data in one slide

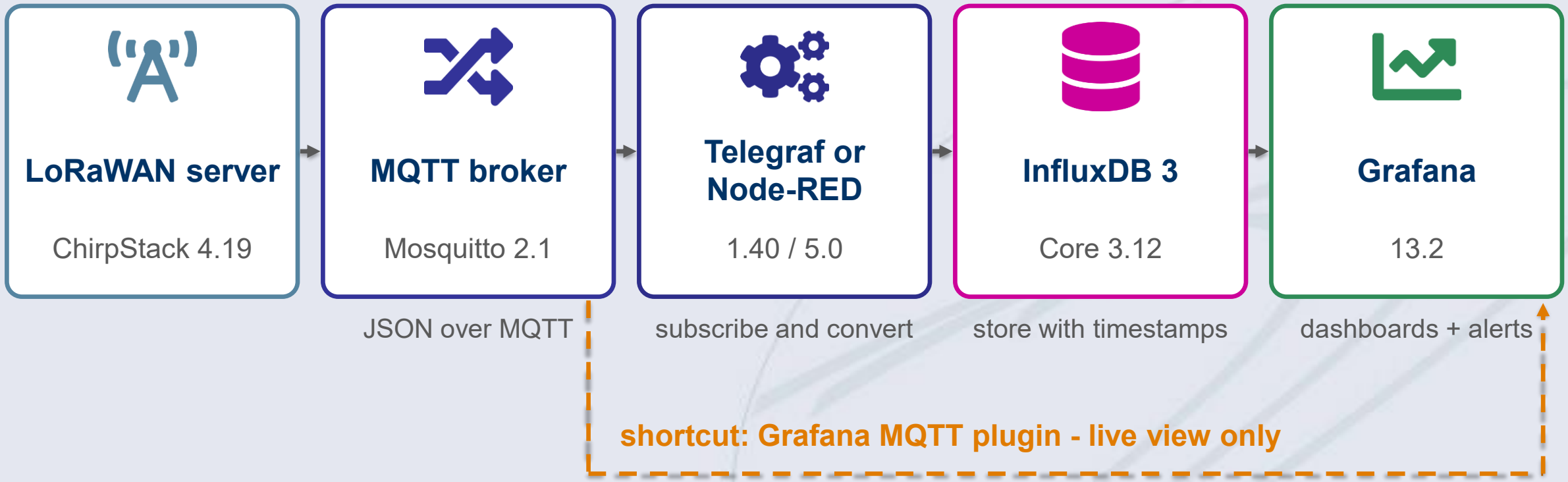
	tags: who and where		field
time	site	sensor	value
10:00:00	pump-1	temperature	23.4
10:00:00	pump-1	battery	3.10
10:10:00	pump-1	temperature	23.9
10:10:00	pump-2	temperature	31.2
10:20:00	pump-1	temperature	24.1

Typical questions

- What is the last value?
- Average per 5 minutes
- Maximum per day
- How long since the last message?

Keep raw data for weeks, keep averages for years (downsampling + retention)

From radio to dashboard



History and alerts need the database in the middle

Where to store sensor data

Option	What it is	Status (Oct 2026)	Good for
InfluxDB 3 Core	open-source time-series DB	stable since Apr 2025	labs, gateways, on-prem
Timestream for InfluxDB	managed InfluxDB on AWS	open to new customers	AWS projects
Timestream LiveAnalytics	older AWS time-series DB	**closed to new customers**	avoid for new work
DynamoDB via IoT rule	AWS NoSQL table	active	cheapest lab store
CloudWatch metrics	AWS monitoring	active	quick charts, no setup

Rule of thumb: store raw data cheaply, and choose a database you can export from

Grafana in five words

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1 **Data source**

InfluxDB, CloudWatch, MQTT

2 **Query**

average temperature per 5 min

3 **Panel**

time series, gauge, stat, table

4 **Dashboard**

one screen, one question

5 **Alert**

rule + who to tell

Grafana today

- Grafana 13.2.3 (29 Sep 2026) - open source, runs anywhere, or as a cloud service
- Grafana 13 (Apr 2026): dashboards as code in Git, dynamic layouts, an AI assistant

Live MQTT straight into Grafana

Grafana MQTT data source plugin 1.3.8 (Sep 2026): subscribes to topics and draws messages as they arrive

Good for

- Class demos - no database at all
- A live wall display
- Checking that data flows

Not for

- History: only while the panel is open
- Alerts: not supported
- MQTT 5: speaks MQTT 3.1.x only

For history and alerts: MQTT -> Telegraf -> InfluxDB -> Grafana

A dashboard for our remote site

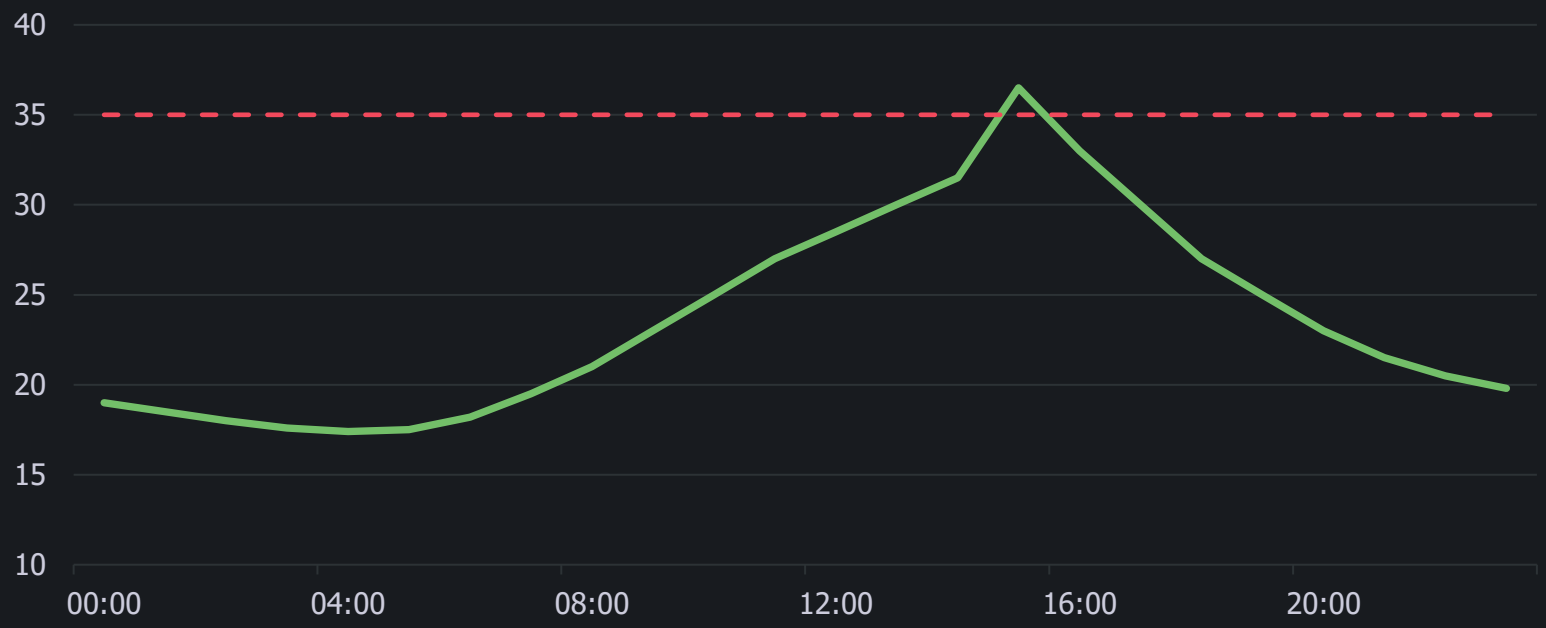
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Remote Site Watch - pump station 1

Last 24 hours

Site status	Last message	Temperature	Battery
ONLINE	42 s ago	23.5 °C	3.1 V

Temperature (°C) and alarm threshold



Door events

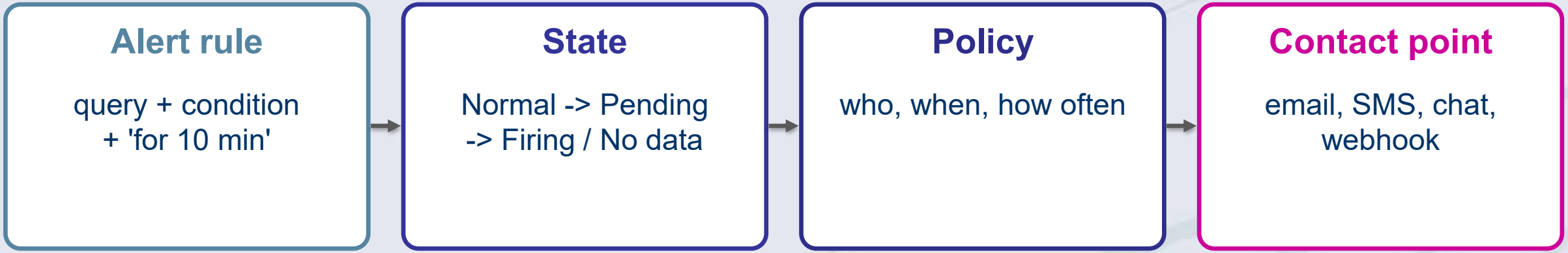
02:14	opened
02:16	closed
11:30	opened
11:41	closed
15:05	temp alarm

Dashboards that operators trust

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- One screen answers one question: is the site OK?
- Status first: traffic lights at the top
- Show 'last seen' - silence is a symptom
- Units and time zone on every panel
- Draw thresholds on the graph
- No 3D, no pie with twelve slices

Alerting: from data to a phone call



Rules for our site

- No data for 15 minutes -> site offline
- Door open for 2 minutes outside working hours
- Temperature above 35 °C for 10 minutes

Alert on symptoms that need a person - not on every spike

Securing the dashboard



Change the default admin password at first login



Never on the Internet without TLS and single sign-on



Operators get the Viewer role, not Editor



Careful with public / shared dashboard links



A read-only database user for every data source

A dashboard can also send commands - protect it like a control system

MAKECODE WARM-UP

30 minutes - getting ready for the Day 4 lab

Microbit.org

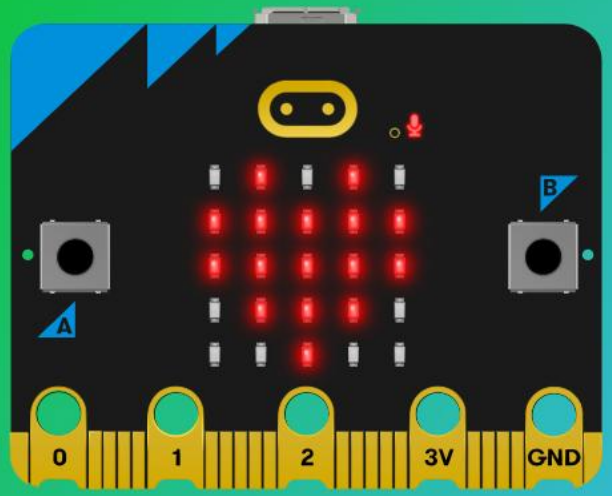
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Language | Help & support | Safety

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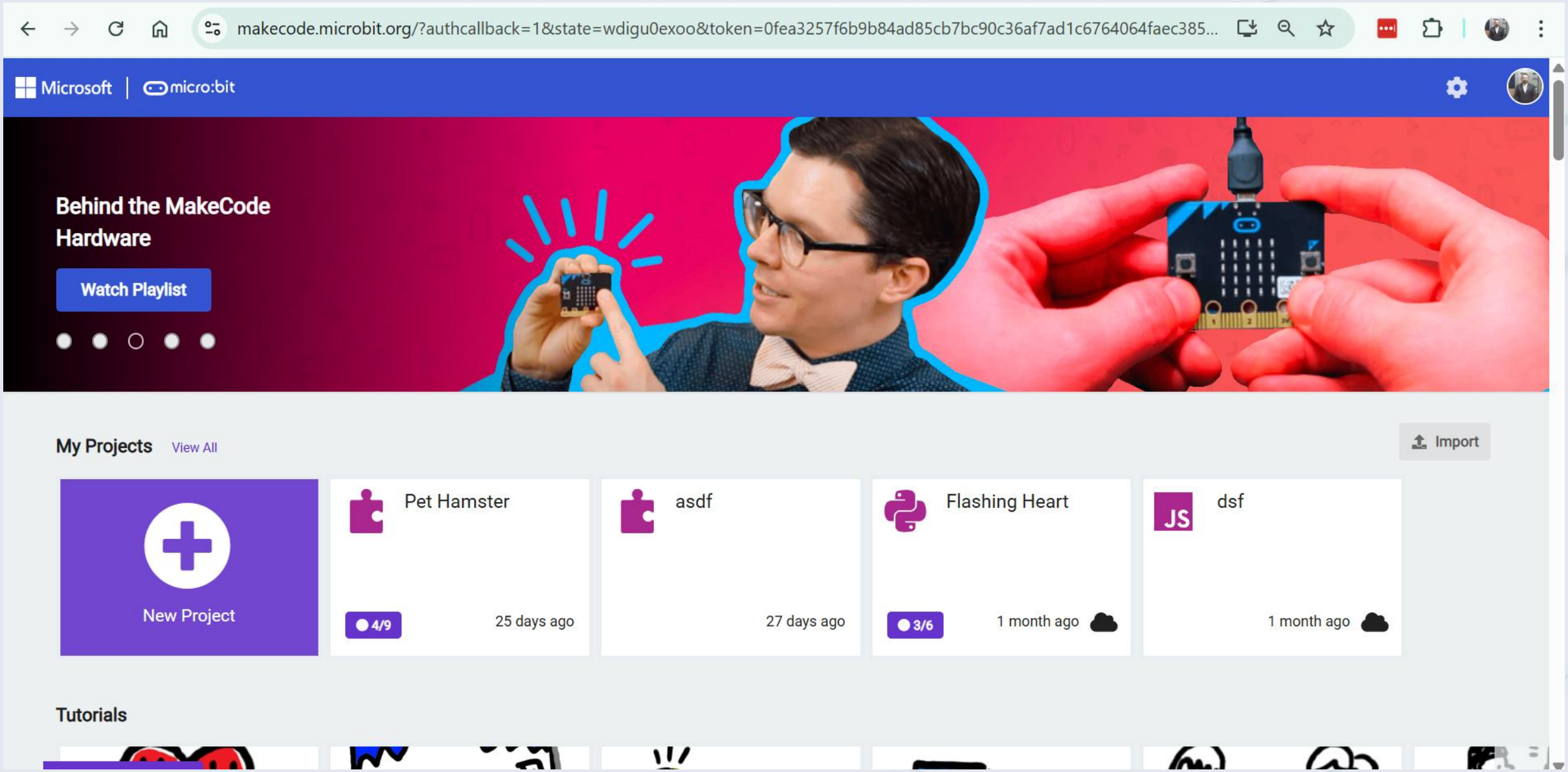
BBC micro:bit



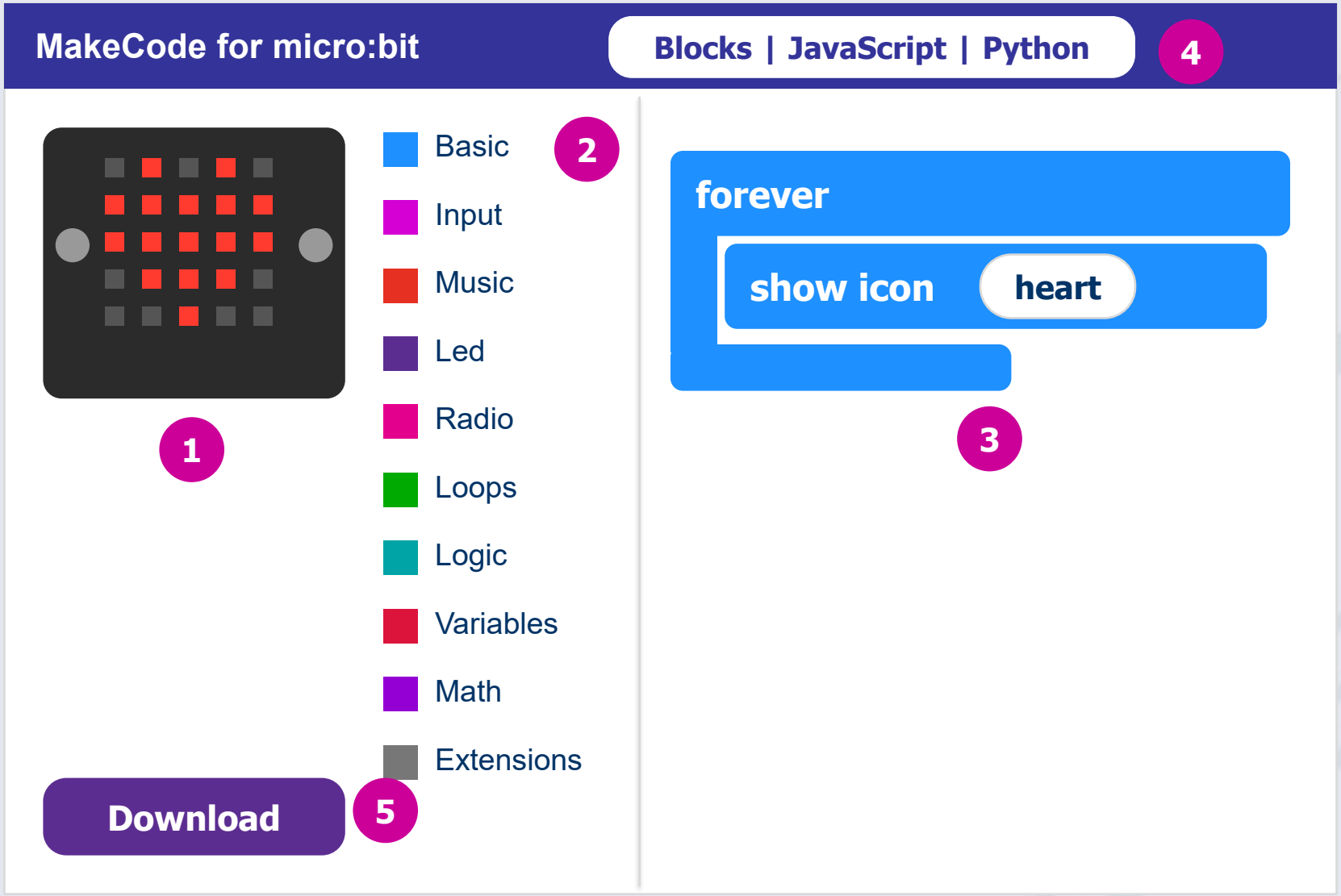
Create | Learn | Code



MakeCode - the micro:bit editor



The MakeCode editor in one slide



1. Simulator: test without hardware
2. Toolbox: drag blocks from here
3. Workspace: your program
4. Same program as JavaScript or Python
5. Download: flash over USB (WebUSB)

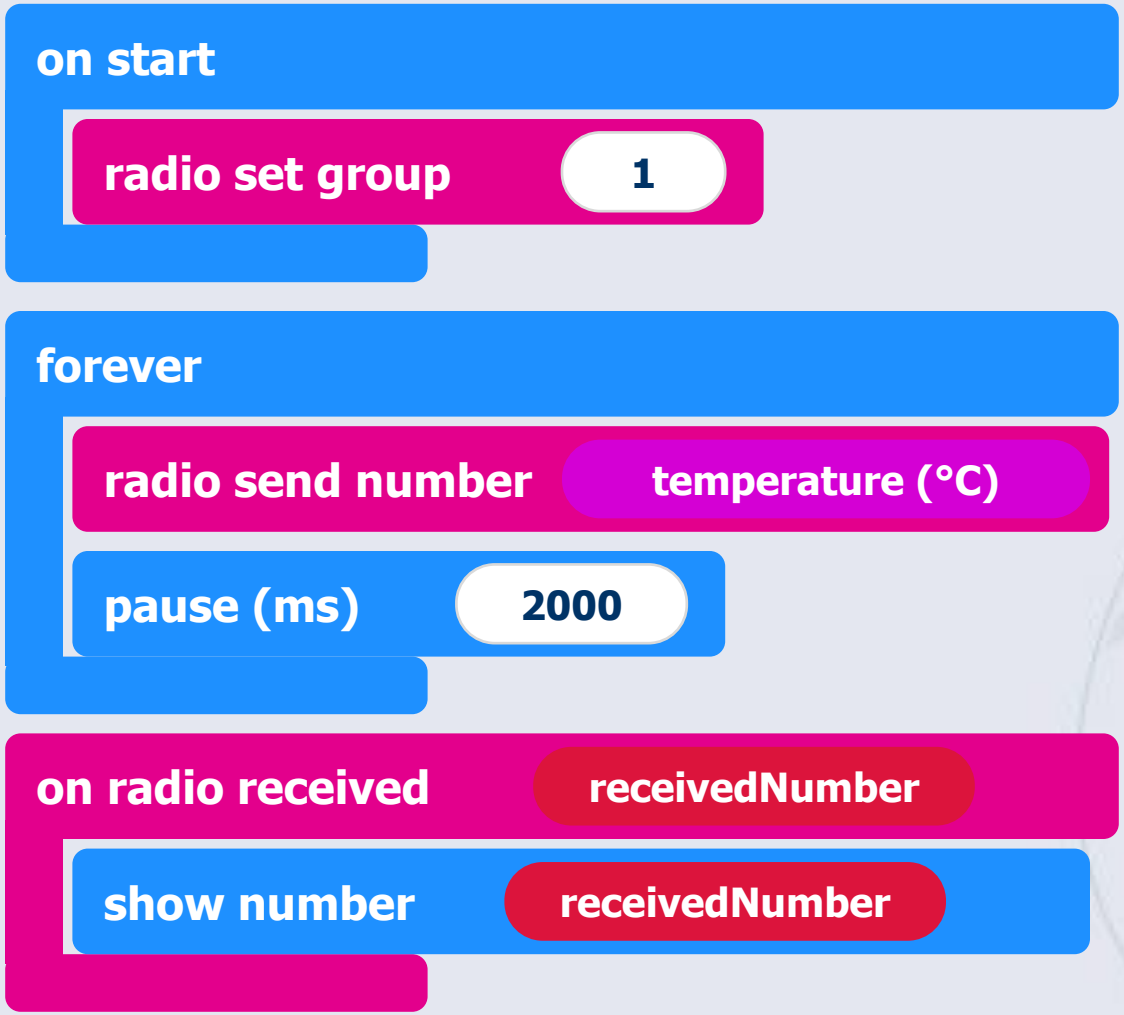
Exercise 1: temperature on the LEDs




```
basic.forever(function () {
    basic.showNumber(input.temperature())
    basic.pause(1000)
})
```

1. Build it from the Basic and Input menus
2. Test it in the simulator
3. Download it to the micro:bit
4. Warm the chip with your finger - does it change?

Exercise 2: send it by radio




```
radio.setGroup(1)
basic.forever(function () {
    radio.sendNumber(input.temperature())
    basic.pause(2000)
})
radio.onReceivedNumber(function (receivedNumber) {
    basic.showNumber(receivedNumber)
})
```

Group = your team number, or you hear the other teams

The micro:bit radio has no encryption: anyone in the group can listen

Exercise 3: a micro:bit gateway




```
radio.onReceivedNumber(function (receivedNumber) {
    serial.writeValue("temp", receivedNumber)
})
```



In MakeCode press 'Show console' to watch the values arrive

This micro:bit is now a gateway: it hears every sensor in its radio group

Our lab = a mini LoRaWAN

LoRaWAN in the field	Our micro:bit lab
End device: sensor + LoRa radio	micro:bit + radio; Day 4: IoT:bit Wi-Fi
Gateway: hears every device nearby	gateway micro:bit; Day 4: Wi-Fi router
Network server: checks every device	class MQTT gateway: team passwords
Application server + storage	AWS IoT Core: rules, DynamoDB
Uplink JSON on MQTT	telemetry JSON on MQTT
AES-128 keys, OTAA join	**radio: none!** Day 4: passwords, X.509

Same architecture, different radio - Day 4 adds the cloud half

When flashing does not work

Problem	Fix
Download does nothing	Use Chrome or Edge; pair once: 'BBC micro:bit CMSIS-DAP'
No pairing window	Download the .hex file and drag it onto the MICROBIT drive
No MICROBIT drive	Try another USB cable - some only charge
Console stays empty	Serial must be 115200 baud; close other apps on the port
Numbers from other teams	Set your own radio group
Radio and Bluetooth together	Not possible in one MakeCode program

Day 2 in six sentences



LoRaWAN: kilometres on a battery - if messages stay small and rare.



The spreading factor trades range for airtime.



In Israel LoRaWAN uses AS923-4, 917-920 MHz.



Mesh networks - Zigbee, Thread, Matter - cover buildings.



MQTT decouples senders and receivers through a broker.



Grafana shows the data; alerts turn it into action.

Quick quiz

1. SF7 or SF12: which reaches farther - and what does it cost?
2. Which LoRaWAN activation should you use, and why?
3. Does site/+/door match site/haifa/pump-1/door?
4. Which QoS levels does AWS IoT Core support?
5. How does the control room learn that a device died silently?
6. Why can't the Grafana MQTT plugin send alerts?

Thank You!!!